

Disorders of bone,joint,soft tissue

GEBI NEMO

- The functions of bone include:
 - ❖ Mechanical support
 - ❖ Transmission of forces generated by muscles
 - ❖ Protection of viscera
 - ❖ Mineral homeostasis
 - ❖ Providing a niche for the production of blood cells
- Bone matrix is composed of an organic component known as osteoid (35%) and a mineral component (65%) .
- In an adult, the presence of **woven bone** is always abnormal, but it is not specific for any particular bone disease

- Peak bone mass is achieved in early adulthood after the cessation of skeletal growth.
- This set point is determined by many factors, including polymorphisms in the vitamin D and LRP5/6 receptors, nutrition, and physical activity.
- Beginning in the fourth decade, resorption exceeds formation, resulting in a decline in skeletal mass.

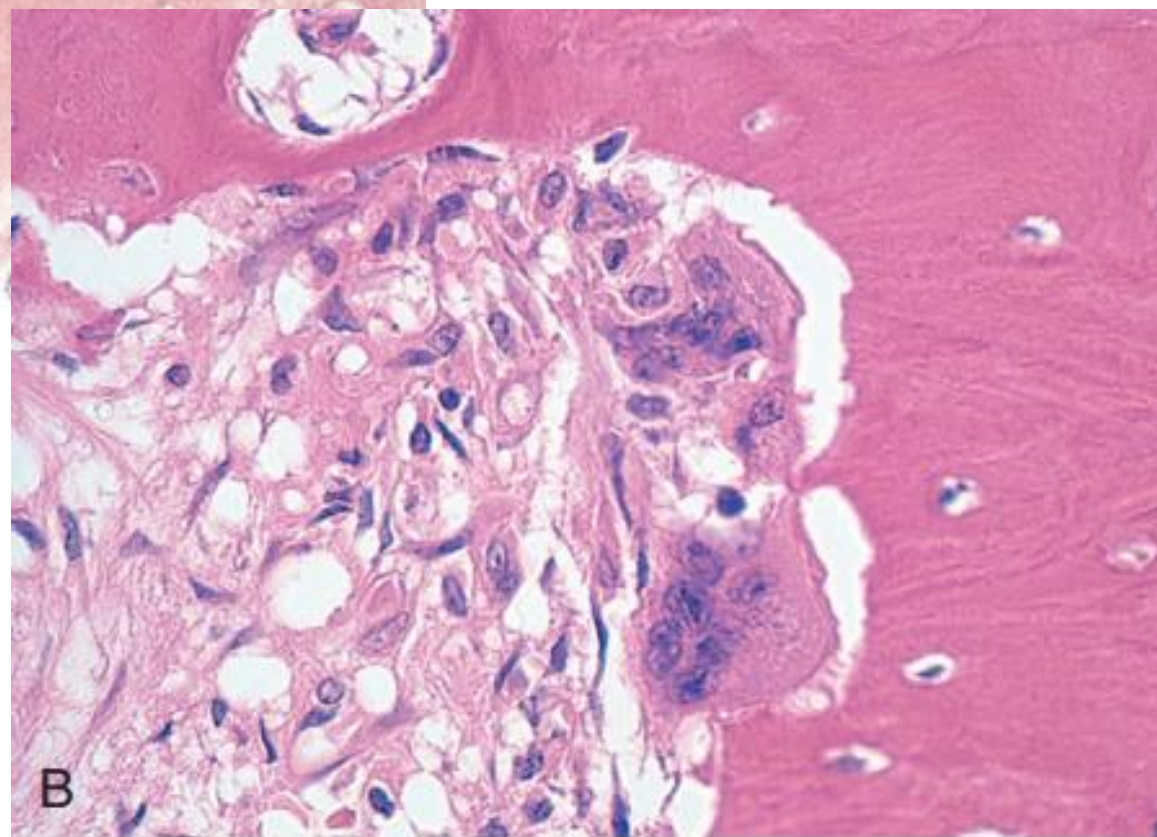
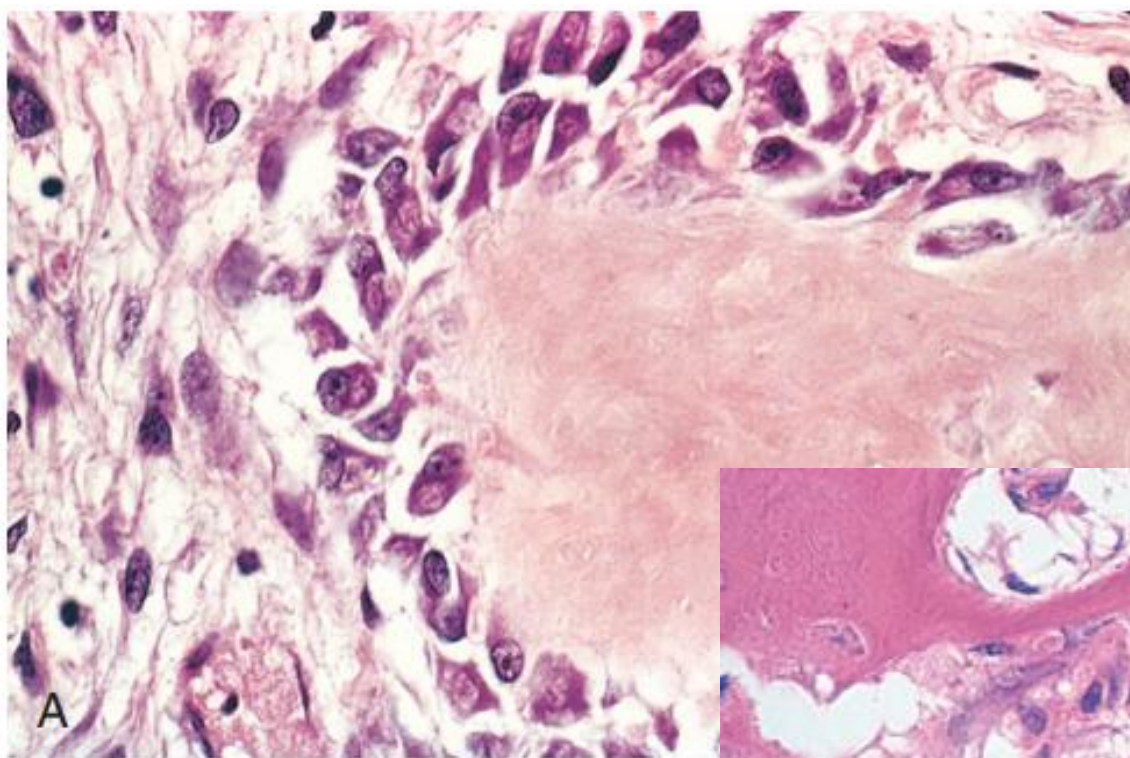
Bone

Structure of Bones

- hard matrix
 - *Made of matrix proteins and mineral*
- The main structural protein-*type I collagen*
- Most of the mineral -in the form of a calcium phosphate complex known as ***hydroxyapatite***

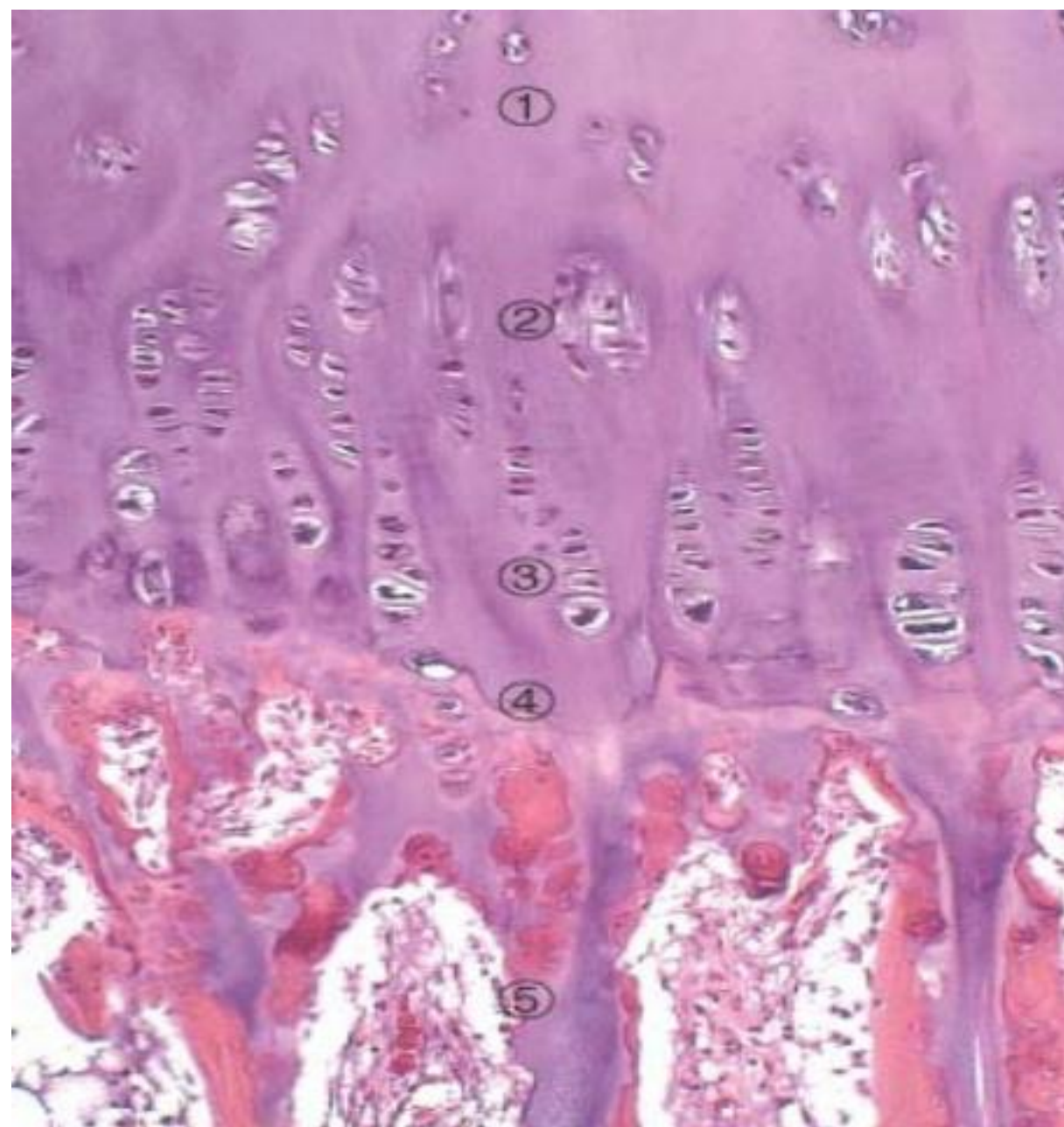
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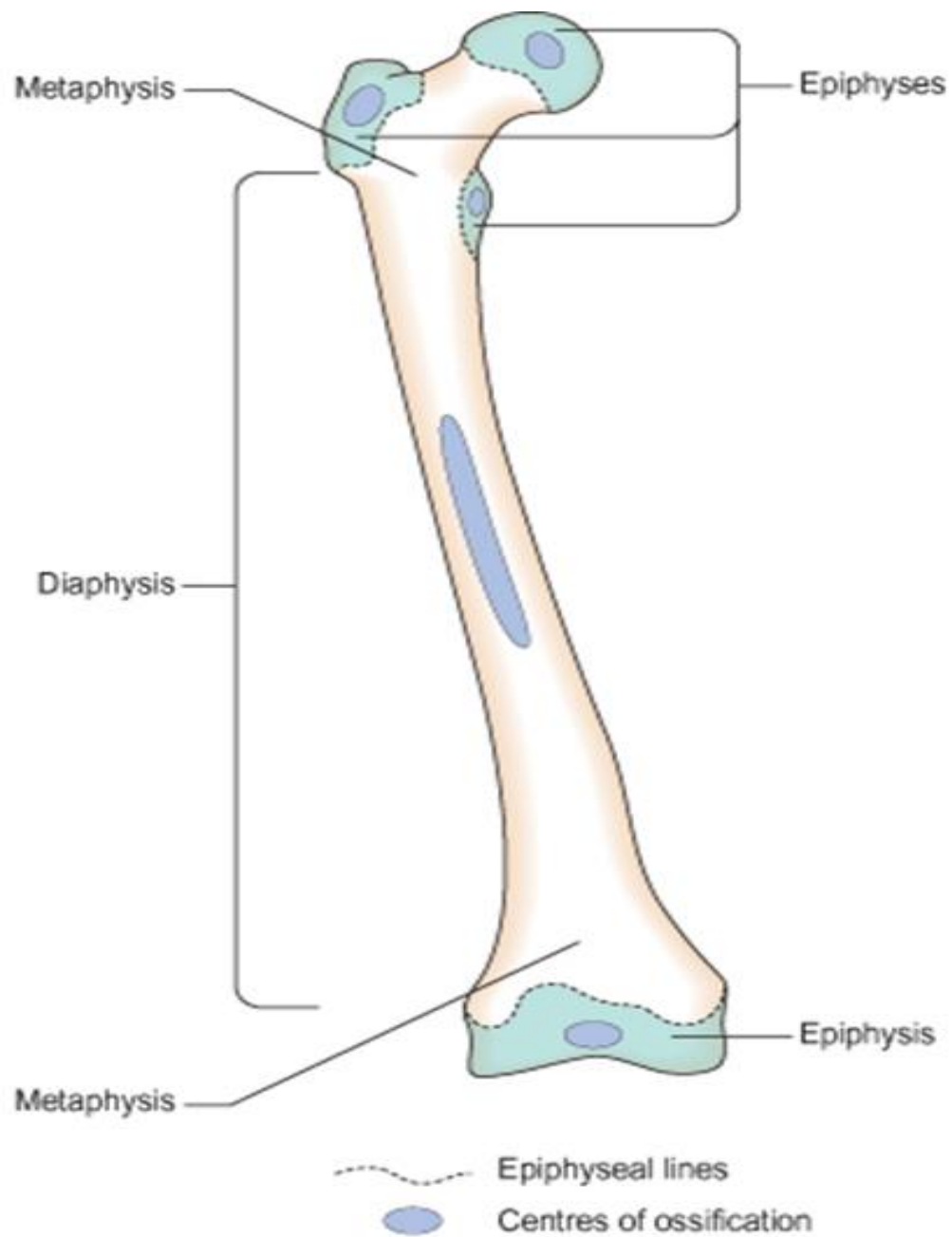
- Bone cells are of two distinct types:
- **Osteoclasts** - bone- resorbing cells
- The osteoblast family :
 - ***osteoblasts***- bone forming cells;
 - ***Osteocytes*** - form an interconnecting network throughout bone matrix
- All are involved in bone remodeling



Cont..

- During development, bone is formed either
 - directly in connective tissue, as in the skull (*intramembranous ossification*) or
 - on pre-existing cartilage, as in the limb bones (*endochondral ossification*)





CONGENITAL AND HEREDITARY DISEASES OF BONE

Achondroplasia

- The disease is caused by gain-of-function mutations in fibroblast growth factor receptor 3 (FGFR3) characterized by impaired maturation of cartilage in the developing growth plate
- It is a major cause of *dwarfism*
- the most common skeletal dysplasia
- not associated with changes in longevity, intelligence, or reproductive status.

Cont..

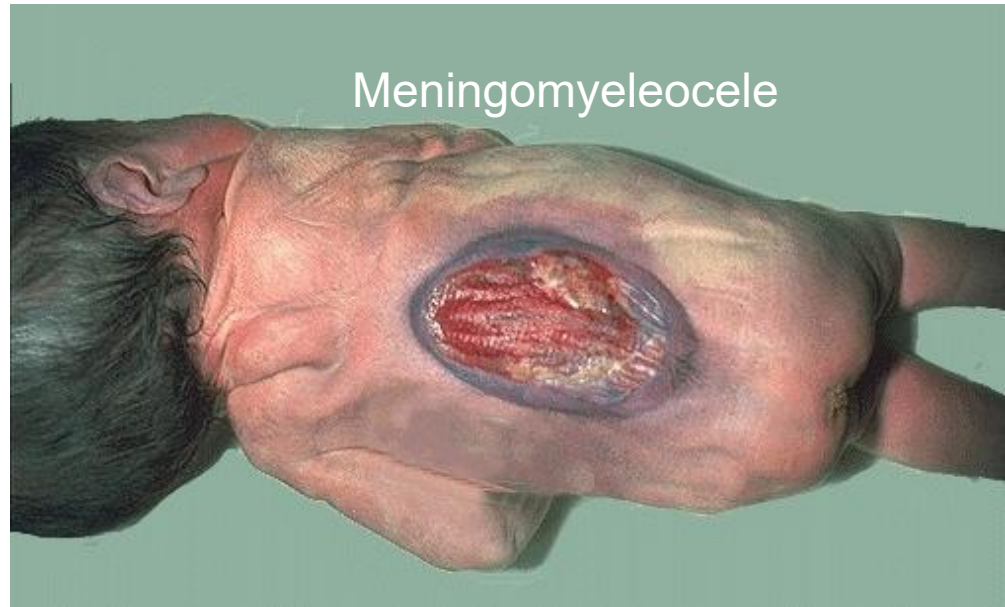
- It affects all bones formed from cartilage
- **normal** head and trunk size, and **disproportionately** short but well-muscled arms and legs
- The face usually has a large forehead, prominent supraorbital ridges, and deep-set root of the nose

Cont..

Craniorachischisis

- *failure of closure of the vertebral column and skull*

➔ ➔ *meningomyelocele or myeloencephalocele*



Cont..

Osteogenesis Imperfecta

- Weak "brittle bone disease"
- the most common inherited disorder of connective tissue
- A group of hereditary conditions characterized by abnormal development of type I collagen ,normally present in skin, joints, and eyes, and it is a major component of **normal osteoid**
- The fundamental abnormality in OI is too little bone, resulting in extreme skeletal fragility
- It may cause multiple bone fractures and fetal demise in utero or shortly after birth.
- Blue sclerae
- Hearing loss
- Small misshapen teeth



Cont..

Osteopetrosis

- "marble bone disease"
- Can be autosomal recessive or dominant
- Characterized by reduced bone resorption and diffuse symmetric skeletal sclerosis resulting from impaired formation or function of osteoclasts.
- Most of the mutations underlying osteopetrosis interfere with the process of acidification of the osteoclast resorption pit, which is required for the dissolution of calcium hydroxyapatite within the matrix.
- These include autosomal recessive mutations in the enzyme carbonic anhydrase 2 (CA2), or mutations in the *TCIRG1* gene, which encodes a component of a vacuolar ATPase that helps to maintain acidic pH in osteoclasts, which is essential for breaking down bone.
- Clinical features
 - increased fractures,
 - anemia, thrombocytopenia, pancytopenia
 - nerve root compression with cranial nerve palsies
 - Vision and hearing impairment
 - Hydrocephalus
 - Renal tubular acidosis
 - Treatment is BM transplant

Osteoporosis and Acquired metabolic diseases

Osteoporosis

- reduction in bone mass in the presence of normal mineralization
- The hallmark of osteoporosis is histologically normal bone that is decreased in quantity
- diagnosed by radiological assessment of bone mineral density

Categories of generalized osteoporosis

Primary
Postmenopausal
Senile
Secondary
Endocrine Disorders
Hyperparathyroidism
Hypo or hyperthyroidism
Hypogonadism
Pituitary tumors
Diabetes, type I
Addison disease
Neoplasia
Multiple myeloma
Carcinomatosis

Gastrointestinal Disorders
Malnutrition
Malabsorption
Hepatic insufficiency
Vitamin C, D deficiencies
Idiopathic disease
Drugs
Anticoagulants
Chemotherapy
Corticosteroids
Anticonvulsants
Alcohol
Miscellaneous
Osteogenesis imperfecta
Immobilization
Pulmonary disease
Homocystinuria
Anemia

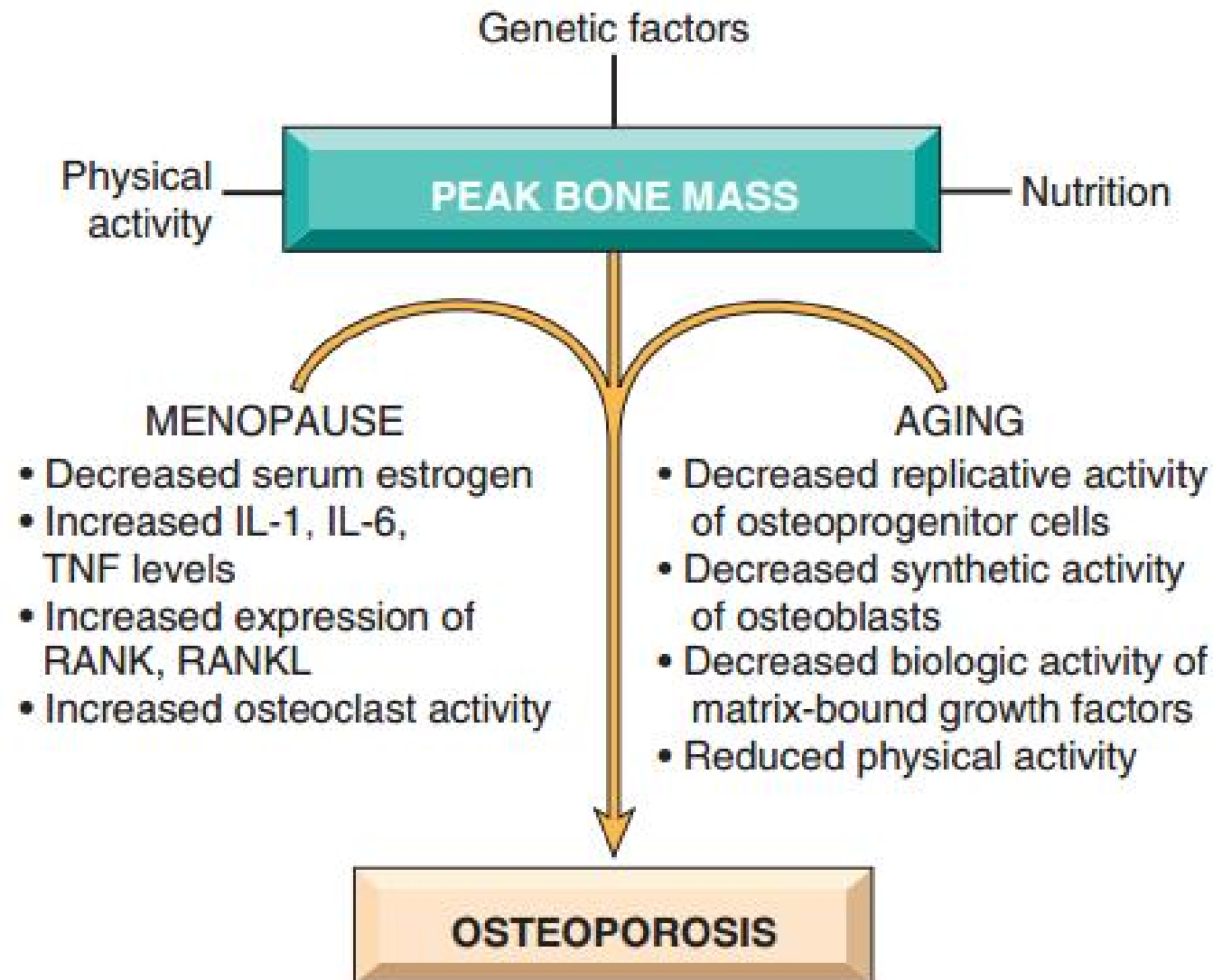


Figure 20–4 Pathophysiology of postmenopausal and senile osteoporosis (see text).

Cont..

Pathogenesis

- is caused by a loss of coupling in the bone remodeling process → net loss of bone volume
- This can be due to
 - *increased bone resorption*
 - *decreased bone formation, or*
 - *Both*
- In contrast to osteomalacia , mineralization of bone is **normal**

Normal vertebra



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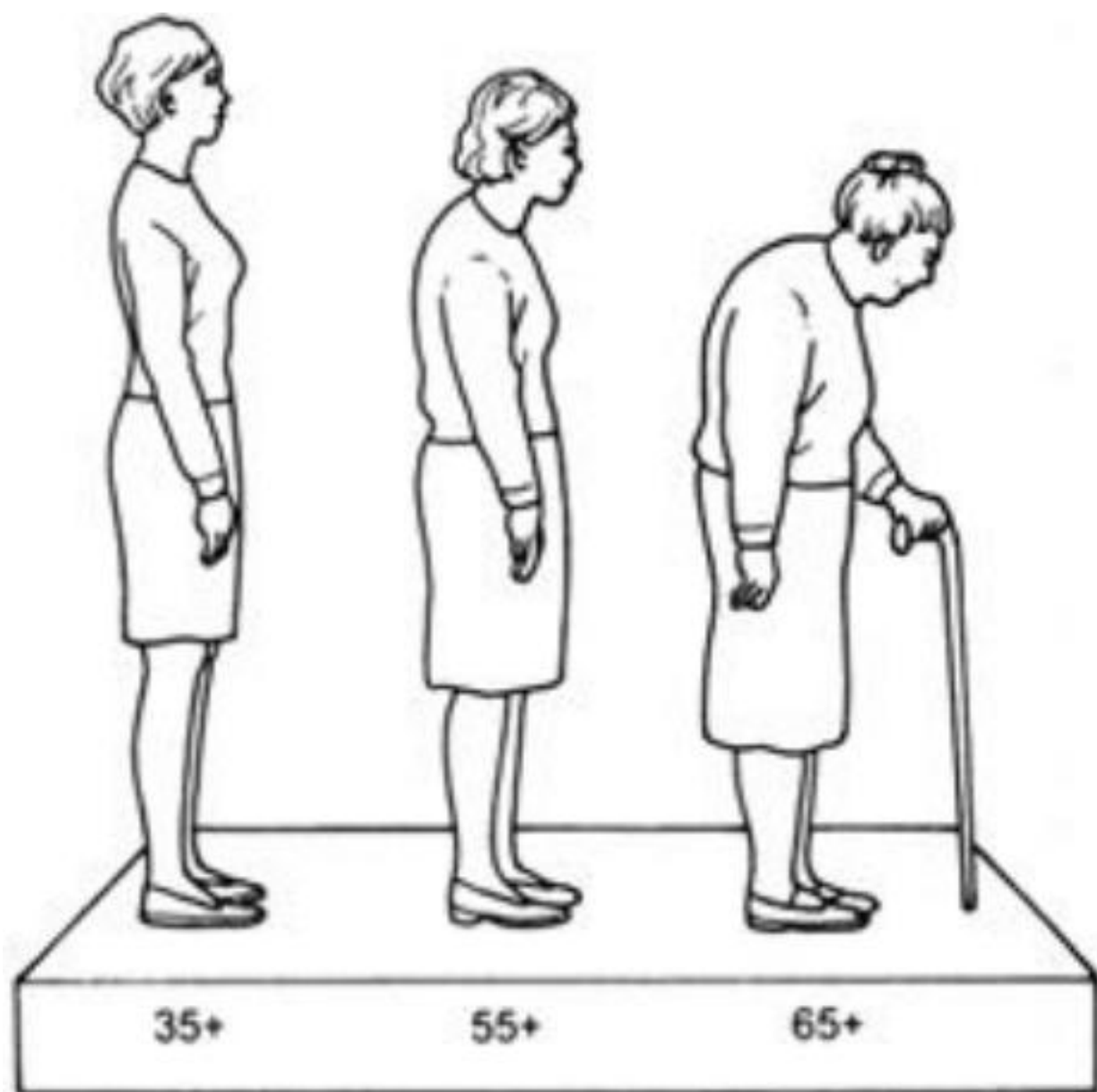
Complications

- The major complications of osteoporosis are:
 - *skeletal deformity*
 - *bone pain (usually due to compression fracture)*
 - *fracture*

Cont..

Clinical

- more common in females than males and white than blacks
- *fragility fracture , progressive loss of height(most common), or stooping deformity (kyphosis or 'dowager's hump')* due to wedge fractures of the vertebral bodies
- Osteoporosis cannot be reliably detected in plain radiographs until 30% to 40% of the bone mass is lost, and measurement of blood levels of calcium, phosphorus, and alkaline phosphatase are not diagnostic
- Corticosteroid is contraindicated
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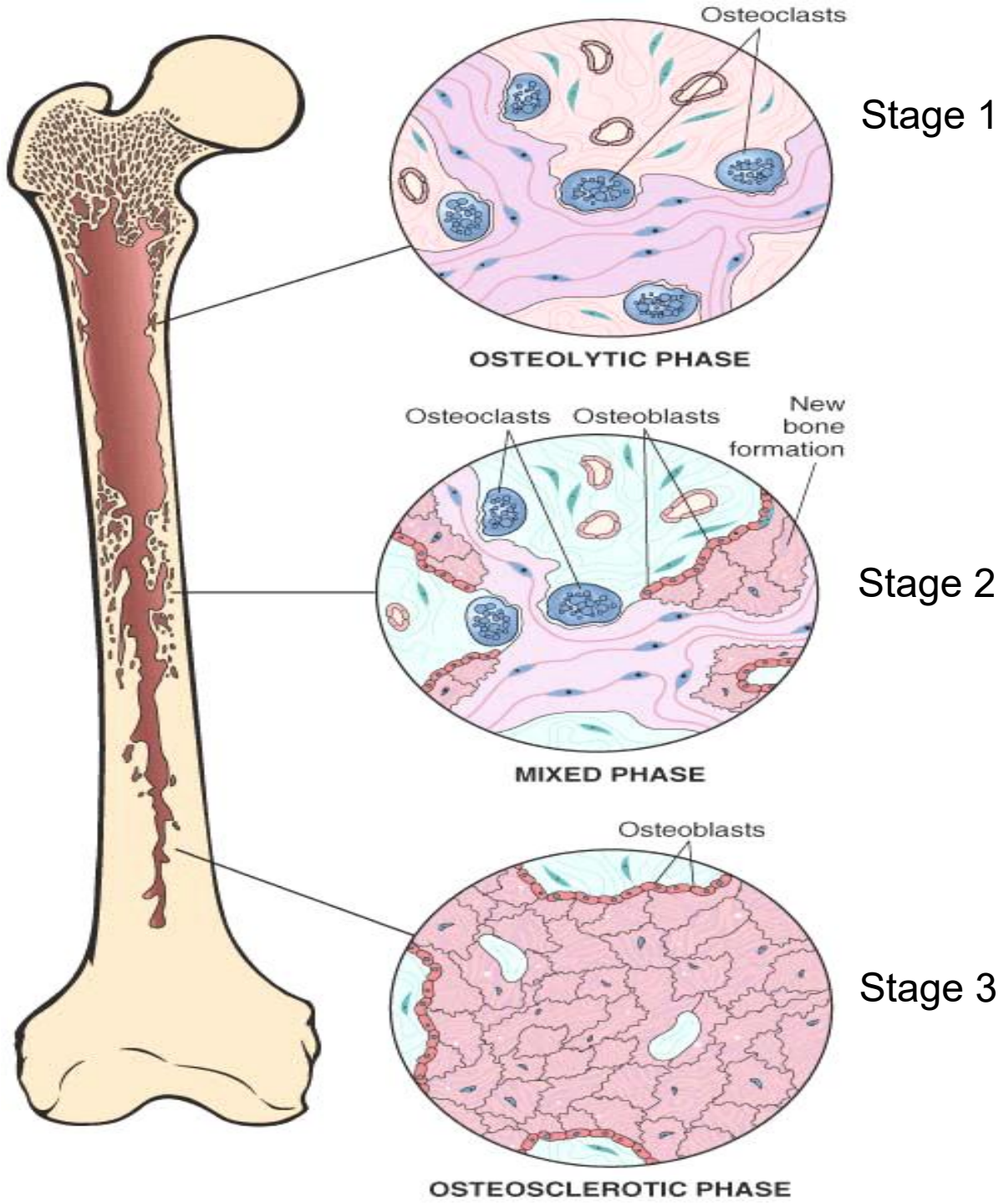


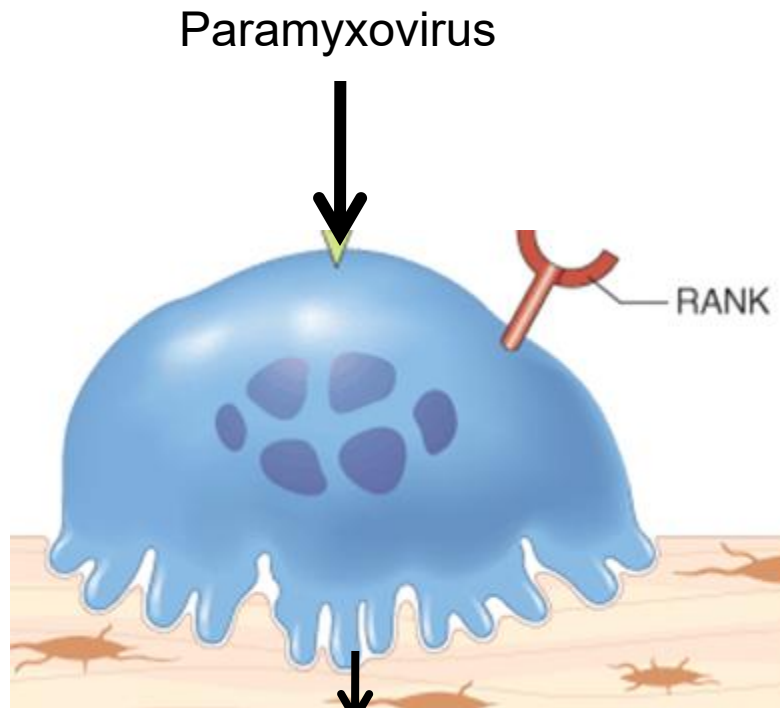
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Paget disease (Osteitis deformans)

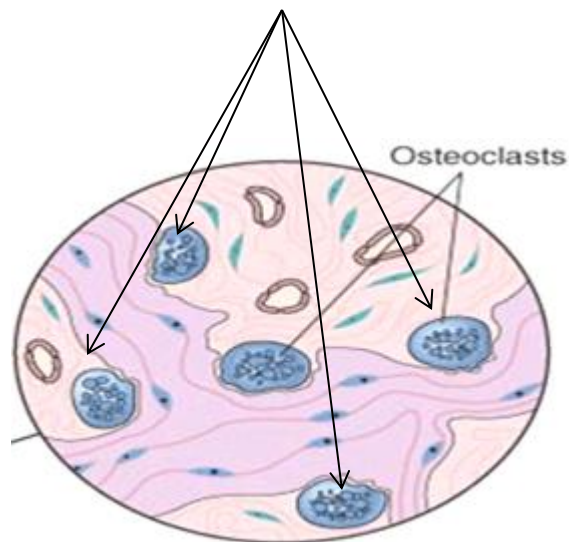
- Characterized by episodes of localized, frenzied osteoclastic activity and bone resorption, followed by exuberant bone formation
- ‘Collage of matrix madness’
- There are three phases in the development of paget disease:

Diagrammatic representation of Paget Disease showing The three phases in the Evolution of the disease





IL-6



paramyxoviruses can induce IL-1
and IL-6 secretion from infected
cells

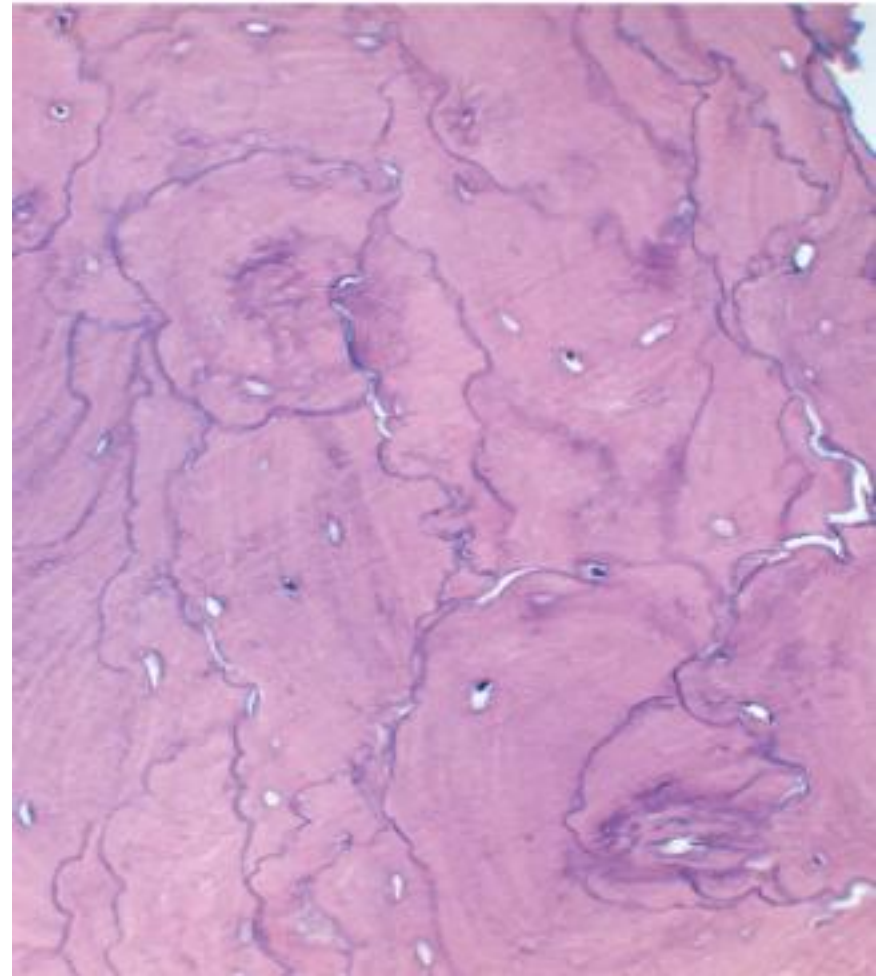
Cont..

- uncommon before 40yrs, but its incidence increases steadily after that time

Morphology

- a solitary lesion (monostotic) or may be multifocal (polyostotic)
- Although any bone may be affected, the **spine, skull, and pelvic bones** are especially common sites of involvement

- Because the bone formation occurs in an erratic pattern, areas of new bone are juxtaposed in a random mosaic pattern, giving the appearance of a jigsaw puzzle



Cont..

Clinical features

- Usually asymptomatic
- High-output congestive heart failure
- Headache, enlargement of the head, visual disturbances, and deafness, all caused by deformity of the bones of the skull and impingement on cranial nerves
- Back pain
- Many affected individuals have elevated serum alkaline phosphatase levels but normal serum calcium and phosphorus.

Cont..

- Transverse fractures of long bones (chalkstick fracture)
- In about 1% of the cases osteosarcoma develops

Cont..

Rickets and osteomalacia

- Characterized by deficient mineralization of the organic matrix of the skeleton
- Rickets - in children
- Osteomalacia -in adults
- Causes include:
 - dietary deficiency of vitamin D
 - deficiency of vitamin D metabolites
 - intestinal Malabsorption
 - renal disease
- Malabsorption of calcium and phosphate from the intestine is the commonest cause of osteomalacia in adults

Cont..

Diagnosis

- The characteristic clinical deformities of rickets include:
 - bowing of the long bones of the leg
 - pronounced swelling at the costochondral junctions
 - cartilaginous proliferation without calcification
 - flattening or 'bossing' of the skull
- Inadequate mineralization of bone reduces its normal strength and allows deformities to develop



Cont..

- Characteristic pathological feature in adults with osteomalacia is spontaneous incomplete fractures
- The main symptoms are bone pain and tenderness, and weakness of proximal limb muscles
- **Serum calcium levels may be reduced and serum alkaline phosphatase is increased**

Infections

Osteomyelitis

- Inflammation of bone and marrow cavity
- Routs of organism entry to the bone:
 1. hematogenous -commonest
 2. direct extension from infected joint or soft tissue
 3. Direct implantation after compound fractures or orthopedic surgical procedures
- *Etiology*
 - *Staphylococcus aureus* -the most common
 - Others pneumococci, *E.coli* and *group B streptococci*
 - *Salmonella*-especially common in sickle cell disease

Cont..

Morphology

- Intense, neutrophilic inflammatory infiltrate at the site of bacterial invasion
- In long bones, the infection spreads through the cortical bone and may reach the periosteum, sometimes creating a subperiosteal abscess
- From the subperiosteal area, the infection may spread into adjacent soft tissues to create **draining sinuses**

Cont..

- The location of infection varies with age
 - In children-**metaphyses** of long bones
 - In adults- primarily affects vertebral bodies that remain quite vascular
 - The involved bone becomes necrotic
- In infants, the existence of loose periosteal attachments and connections between the vessels in the metaphysis and epiphysis allows the infection to spread to the epiphysis and joint capsule

Cont..

Chronic osteomyelitis

- Develops as a sequel of acute infection
- Over time-a repair reaction (osteoclast activation, fibroblastic proliferation, and new bone formation)

Cont..

- **Sequestrum**-residual necrotic bone may be resorbed by osteoclastic activity
- **involucrum**-new reactive bone surrounding sequestrum
- When a well-defined rim of sclerotic bone surrounds a residual abscess, the lesion is sometimes designated a **Brodie abscess**
- Chronic osteomyelitis may be complicated by the development of draining sinuses and pathologic fractures

Cont..

- Other complications- **septicemia, acute bacterial arthritis, squamous cell carcinoma, amyloidosis**

Clinical feature

- Initially systemic manifestations such as *fever, malaise, and leukocytosis*
- local pain, swelling, and redness may occur in some adults

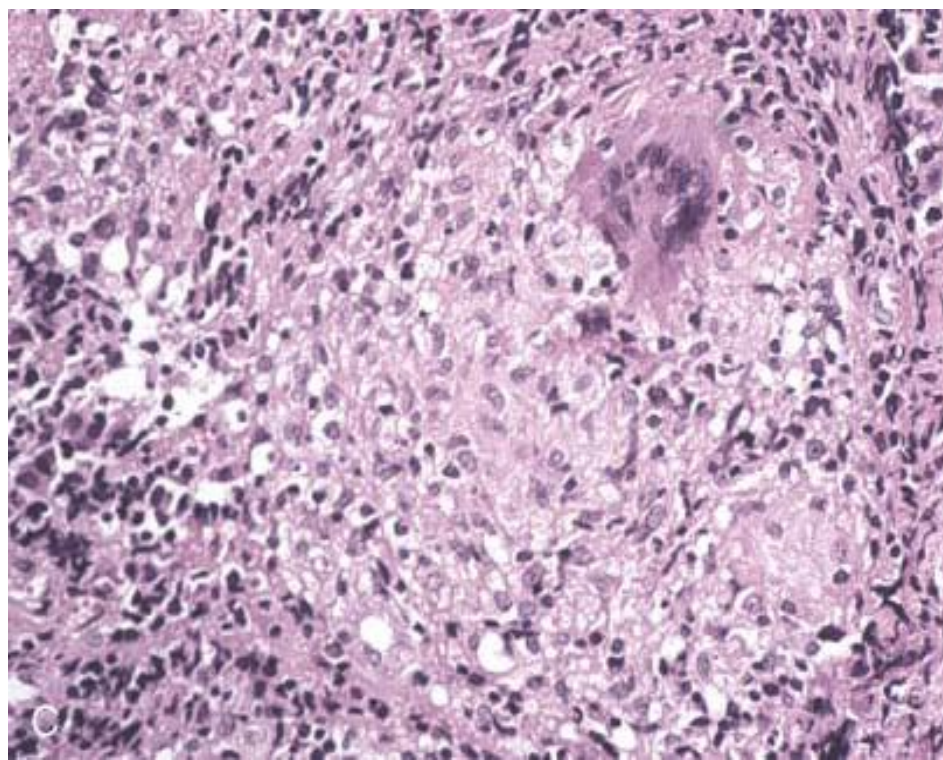
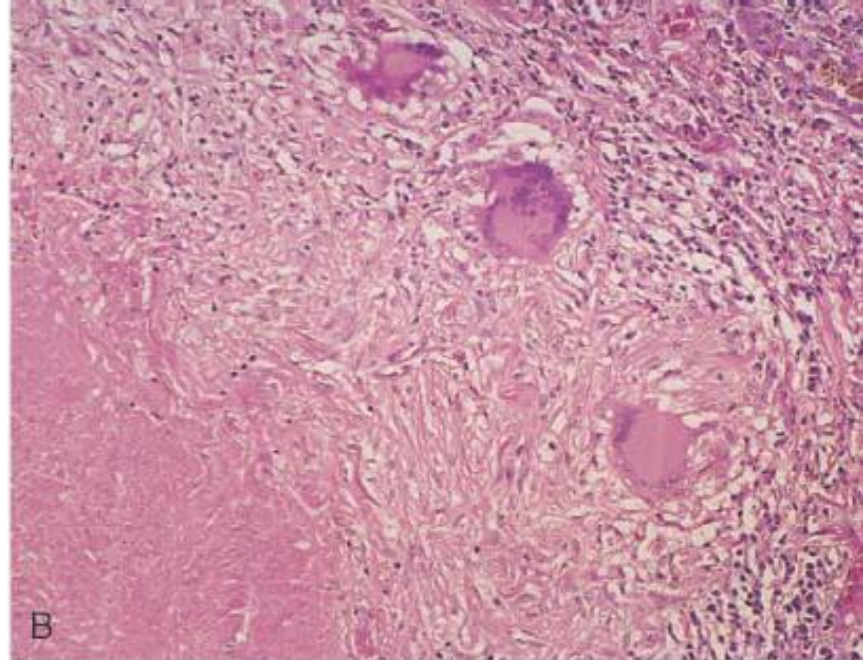
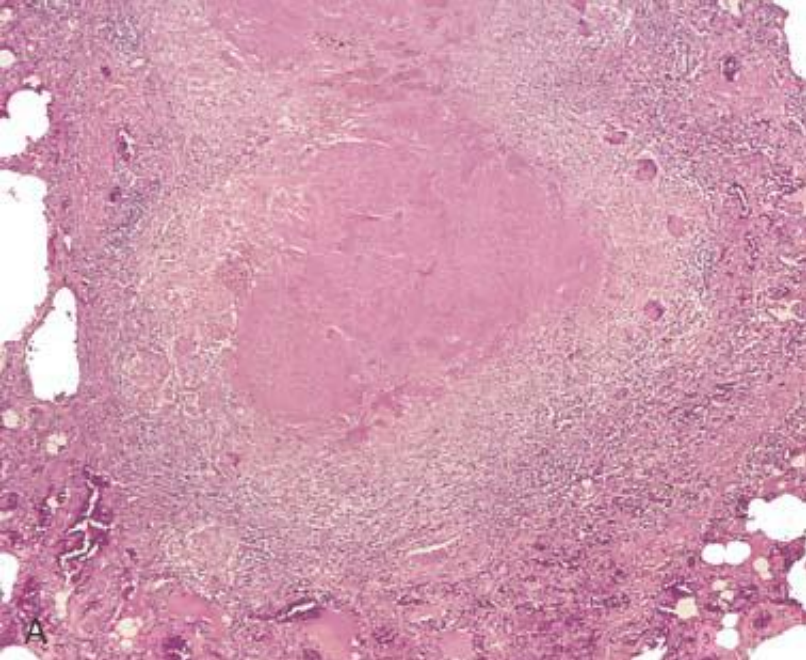
Tuberculous Osteomyelitis

- Hematogenously born
- Rarely direct extension eg. From the lung to the ribs or from the tracheobronchial nodes to the vertebrae
- Bone infection is usually solitary but can be multiple in HIV/AIDS
- The spine (esp. thoracic and lumbar) is the most common site; followed by the knees and hips
- **Pott disease** – in the spine, infection extends through intervertebral discs to involve multiple vertebrae & extends into soft tissues, forming abscess(**psoas abscess**)

Cont..

Clinical feature

- ✓ Pain on motion
- ✓ Swelling
- ✓ Symptom complex of tuberculosis
- ✓ Vertebral deformities



Cont..

Skeletal Syphilis

- Acquired or congenital
- commonly involves- nose, palate, skull & extremities(esp tibia) bones
- Osteochondritis and periostitis
- Spirochetes can demonstrated with silver stains
- Gummata also occur in the bone lesions
- Edematous granulation tissue containing numerous **plasma cells** and necrotic bone

BONE TUMORS

- Benign or malignant
- Malignant- primary(de novo) or secondary
- risk factors- Paget diseases , radiation ,fibrous dysplasia, hereditary (p⁵³ and RB genes)
- The predilection of specific types of tumors for certain age groups and particular anatomic sites provides important diagnostic clues.

Can be

- Bone forming
- Cartilage forming

Or

- Radiographic imaging studies have an important role in diagnosing these lesions.
- In almost all instances biopsy is necessary for definitive diagnosis.
- In older adults, a bone tumor is more likely to be malignant
- Benign tumors greatly outnumber their malignant counterparts

Cont..

Bone-Forming Tumors

- Characterized by the production of osteoid by the tumor cells
- Benign
 - osteoma
 - Osteoid Osteoma
 - Osteoblastoma
- Malignant
 - Osteosarcoma

Osteoma

- Involve skull and facial bones as a exophytic , round mass
- Usually solitary
- Middle age adults
- Multiple in Gardner syndrome
- Histology-composed of **woven and lamellar bone** in a **cortical pattern** with haversian-like symptoms

Cont..

- Slowly growing impinge on brain or eyes or bring cosmetic problems
- Doesn't transform in to osteosarcoma

Osteoid osteoma and Osteoblastoma

- Identical histology but different size ,site and symptoms

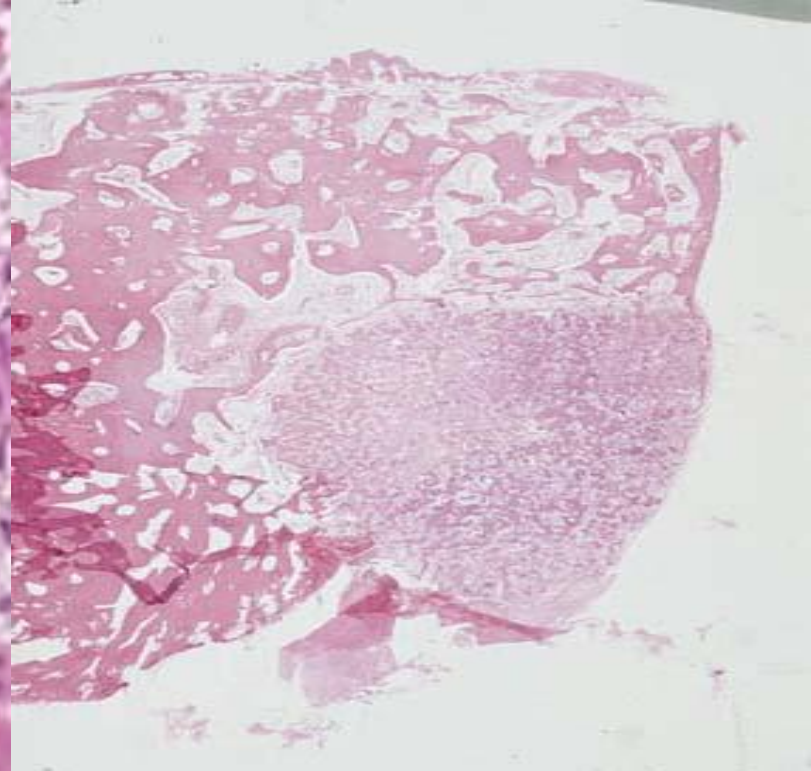
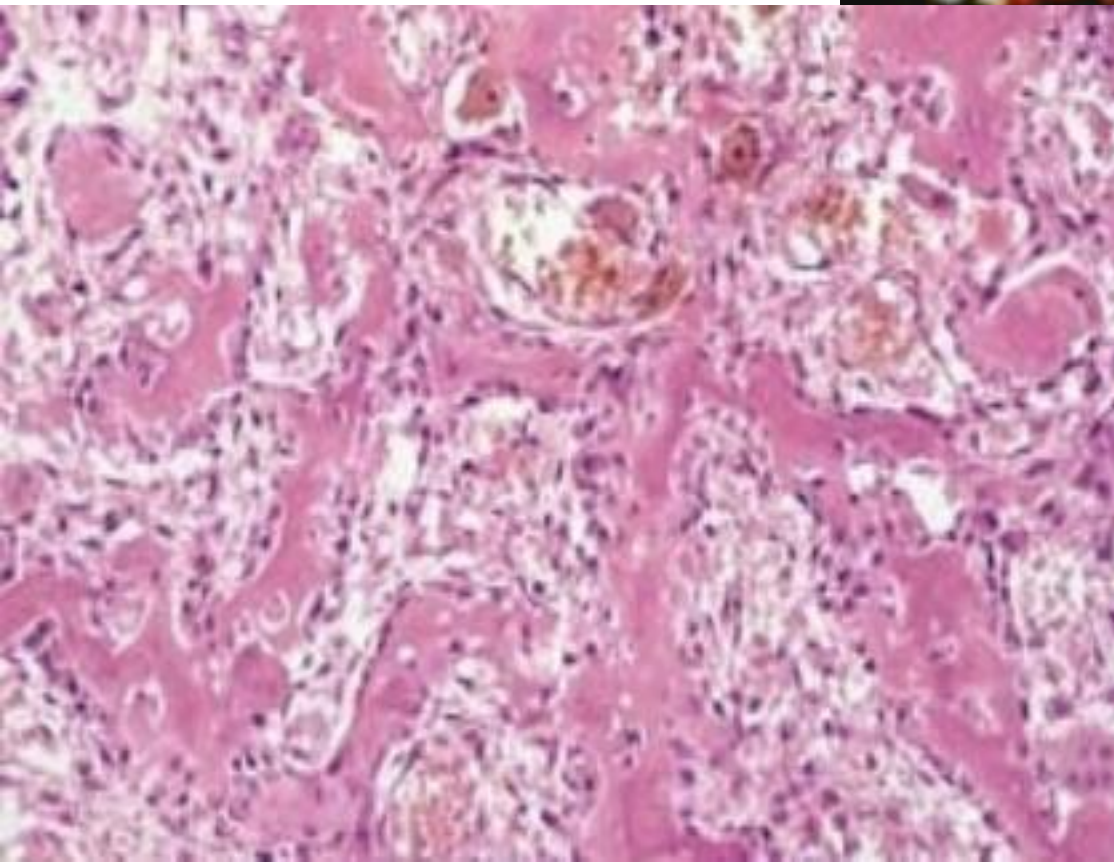
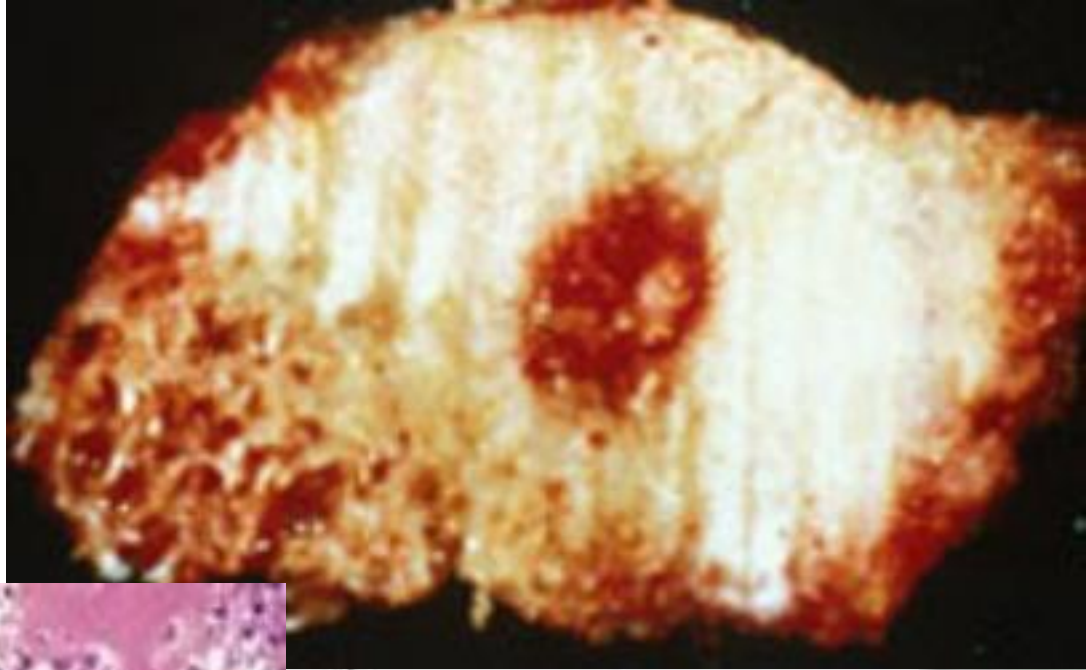
Osteoid osteoma	Osteoblastoma
➤ <i>< 2cm in diameter</i>	➤ <i>Larger</i>
➤ <i>< 25yrs of age</i>	➤ <i>Relatively older</i>
➤ <i>Affect appendicular skeleton esp. femur & tibia</i>	➤ <i>Affect mostly the spine</i>
➤ <i>Very painful (nocturnal) which responds to aspirin</i>	➤ <i>Dull achy pain which doesnot respond to aspirin</i>
➤ <i>Elicits tremendous amount of reactive bone</i>	➤ <i>Less reactive bone</i>

Cont..

Morphology

- Gritty tan hemorrhagic tissue
- Well circumscribed, trabeculae of woven bone lined by osteoblasts, well vascularised, hemorrhagic stroma , surrounding marked reactive bone leaving the tumor as a central nidus
- Osteoid osteoma is frequently treated by radiofrequency ablation, whereas osteoblastoma is usually curetted or excised.

Central hemorrhagic nidus
surrounded by dense rim of
sclerotic bone

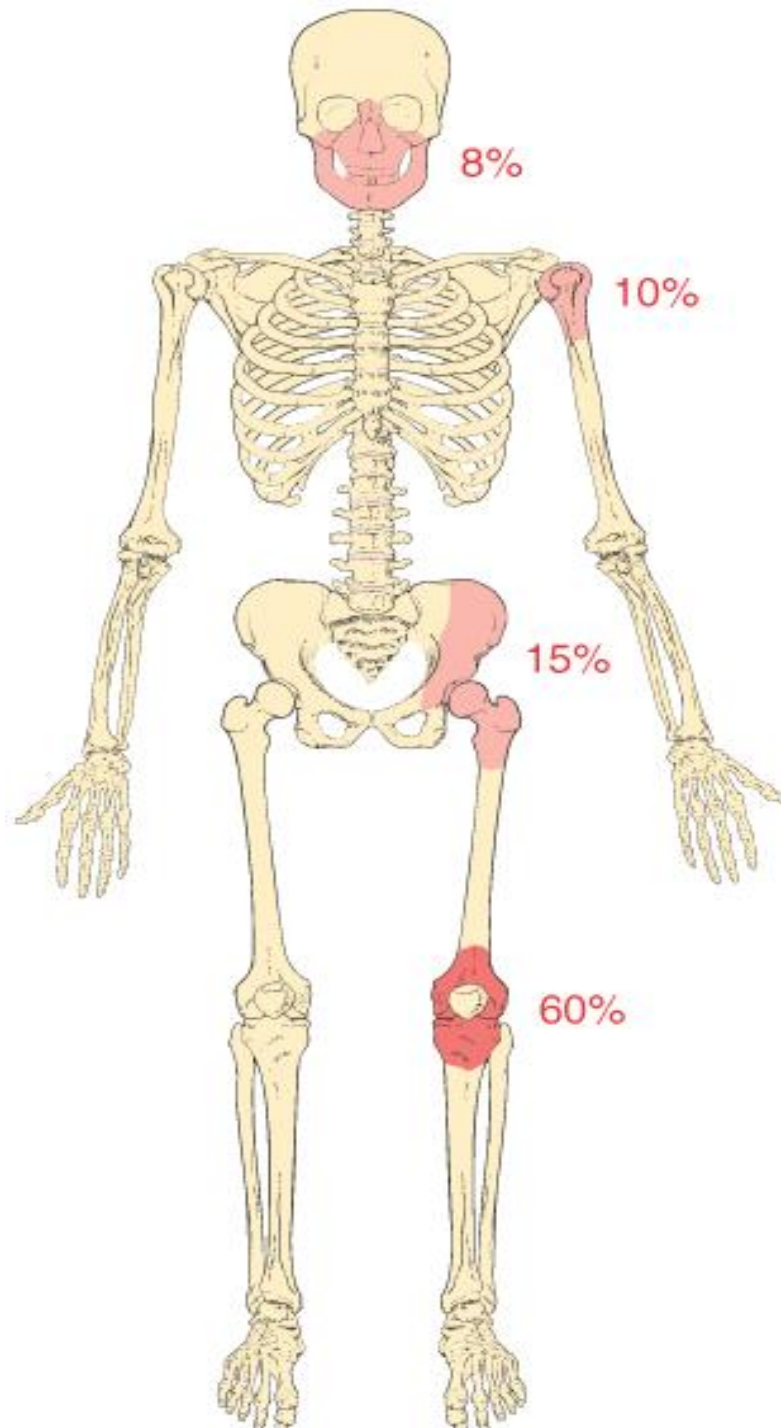


Cont..

Osteosarcoma

A malignant mesenchymal tumor in which the neoplastic cells produce bone matrix

- 75% of cases are < 20yrs of age
- In the elderly it usually follows paget disease, bone infarcts and/or prior irradiation
- Mostly arise in the metaphyseal regions of long bones (60% occur around the knee)



Cont..

- Conventional osteosarcomas are aggressive lesions metastasize through bloodstream early in their course
- The lungs are common sites of metastases
- M > F

Cont..

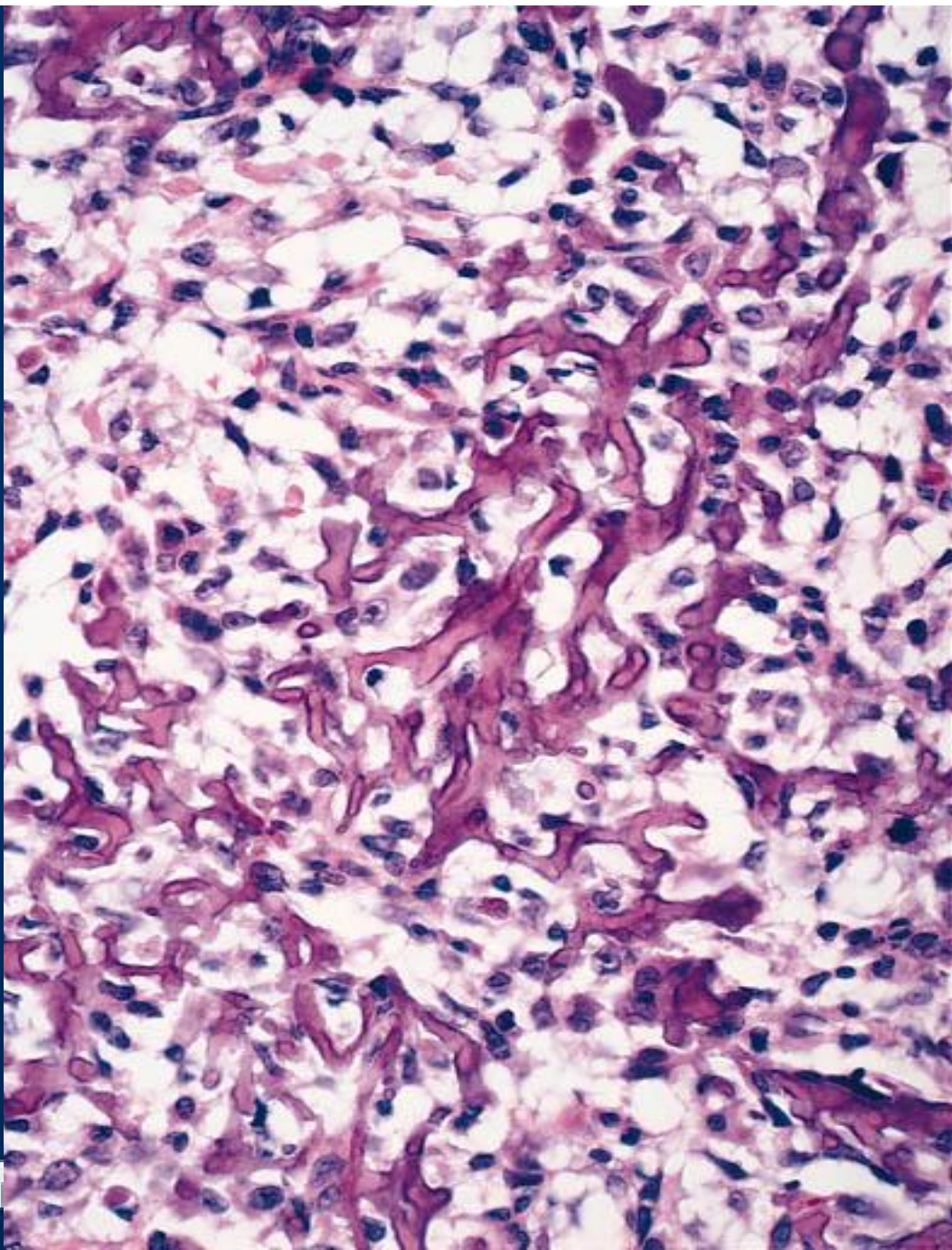
Morphology

- Grossly, osteosarcomas are bulky tumors that are gritty, gray white, and often contain areas of hemorrhage & cystic degeneration.
- The tumors frequently destroy the surrounding cortices and produce soft tissue masses
- Is potentially infiltrating extending in all directions

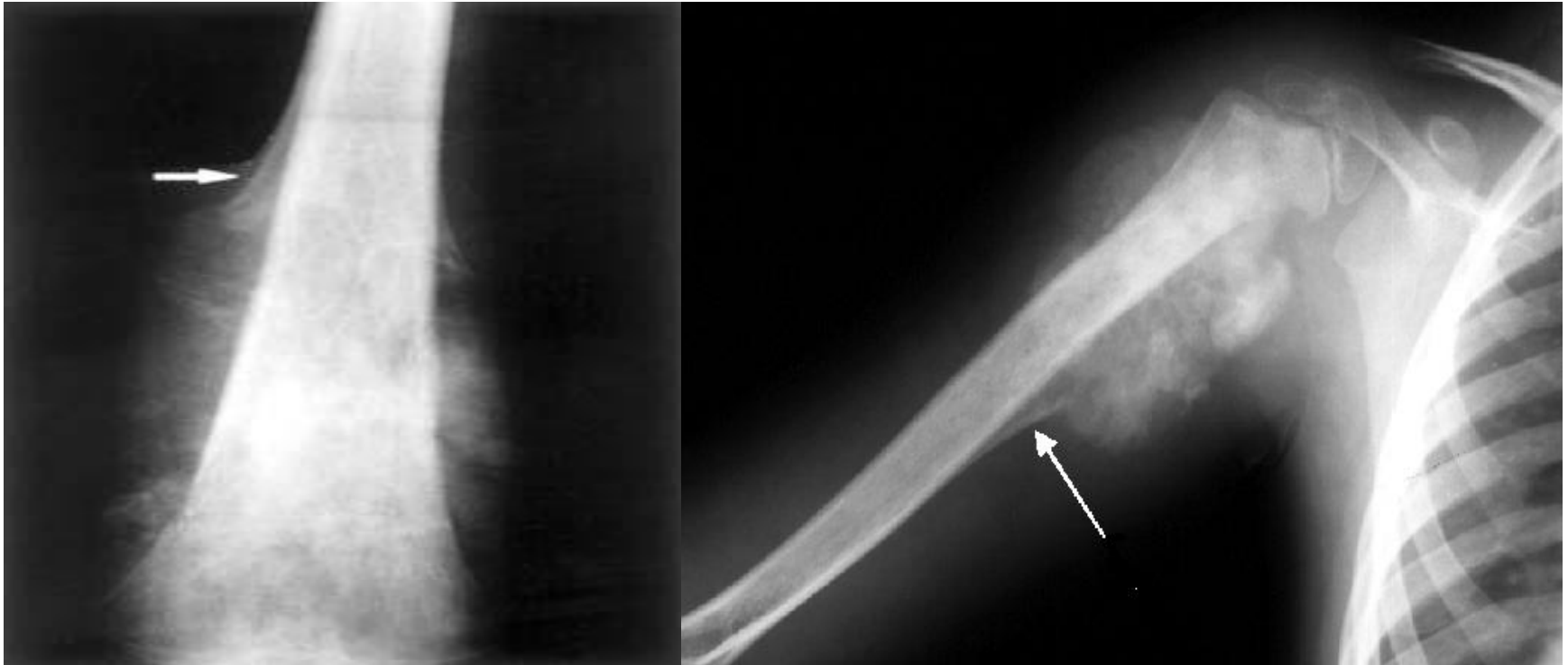
- The neoplastic bone usually has a fine, lacelike configuration but also may be deposited in broad sheets or as primitive trabeculae.
- The tumor cells vary in size and shape (pleomorphic) and frequently have large hyperchromatic nuclei.
- Bizarre tumor giant cells, vascular invasion, and necrosis are common.
- Mitotic activity is high, including abnormal forms (e.g., tripolar mitoses)
- Osteosarcoma is treated with a multimodality approach that consists of (1) neoadjuvant chemotherapy, (2) surgery, and (3) chemotherapy

Cont..

- microscopy– osteoblastic, *chondroblastic, fibroblastic, telangiectatic, small cell and giant variants*
- *The most common variant is that which arises in the metaphysis of long bones; is primary, solitary, intramedullary and poorly differentiated with production of bone matrix*



- The tumor often elevates the periosteum to produce the so-called **Codman triangle** on radiographs



- the hallmark of osteosarcoma is the formation of osteoid by malignant mesenchymal cells!

Cont..

Cartilaginous Tumors

- Benign:
 - Chondroma
 - Osteochondroma
 - Chondroblastoma
 - Chondromyxoid Fibroma
- Malignant:
 - Chondrosarcoma

Cont..

Chondroma (enchondroma)

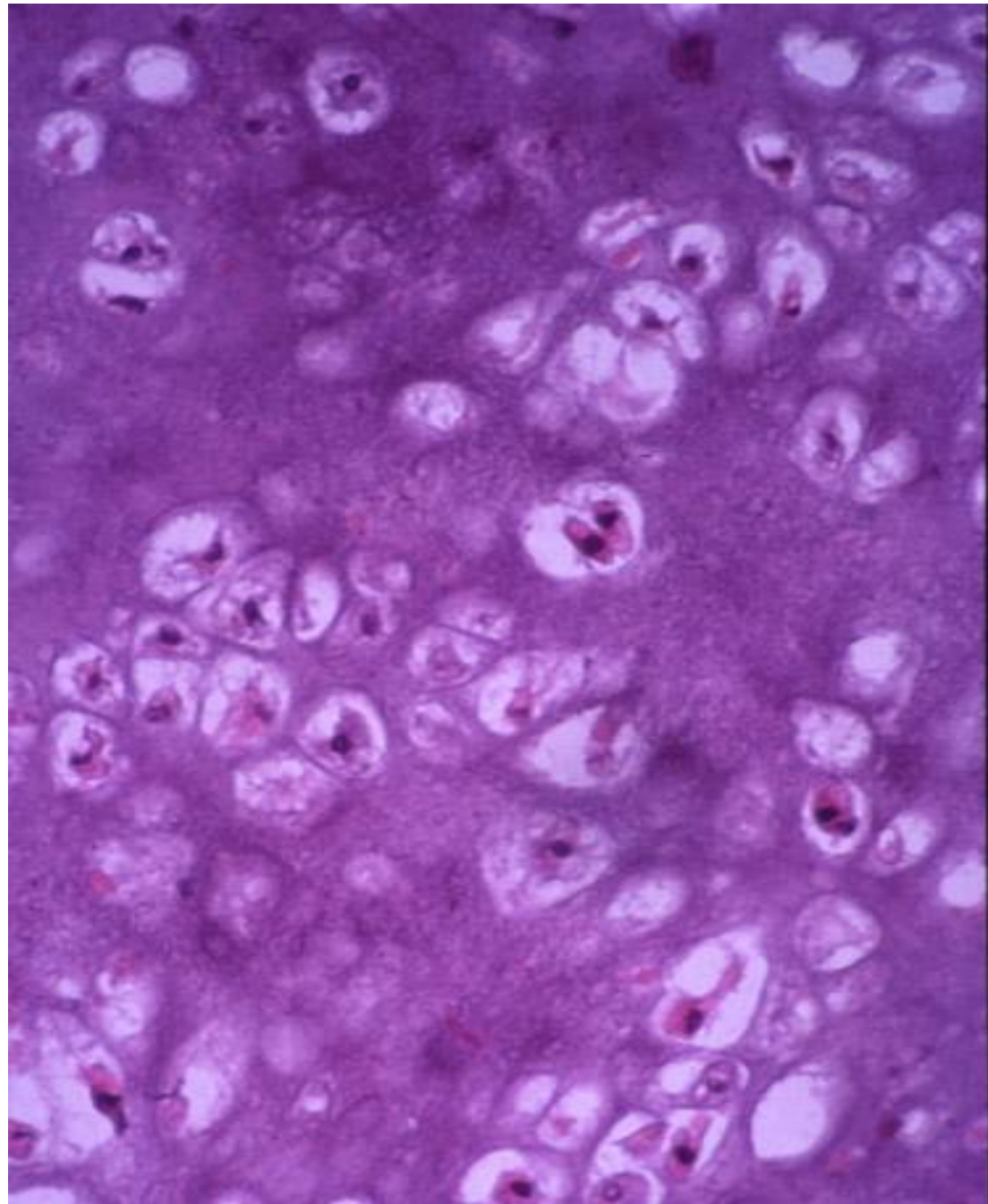
- Benign tumor of mature hyaline cartilage
- Most within bone (enchondroma)
- Most common intraosseous tumors occurring in patients aged between 20 and 50yrs.
- Usually solitary involving the metaphyseal region of tubular bones especially short bones of hand and feet.

Cont..

- Multiple enchondromatoses - Ollier disease, in maffuci's syndrome
- Develop from the rests of cells of the growth plate cartilage and/or genetic alterations in mesenchymal stem cells.



Composed of mature lobules of hyaline cartilage with foci of myxoid degeneration, calcification and endochondral ossification; may be quite cellular



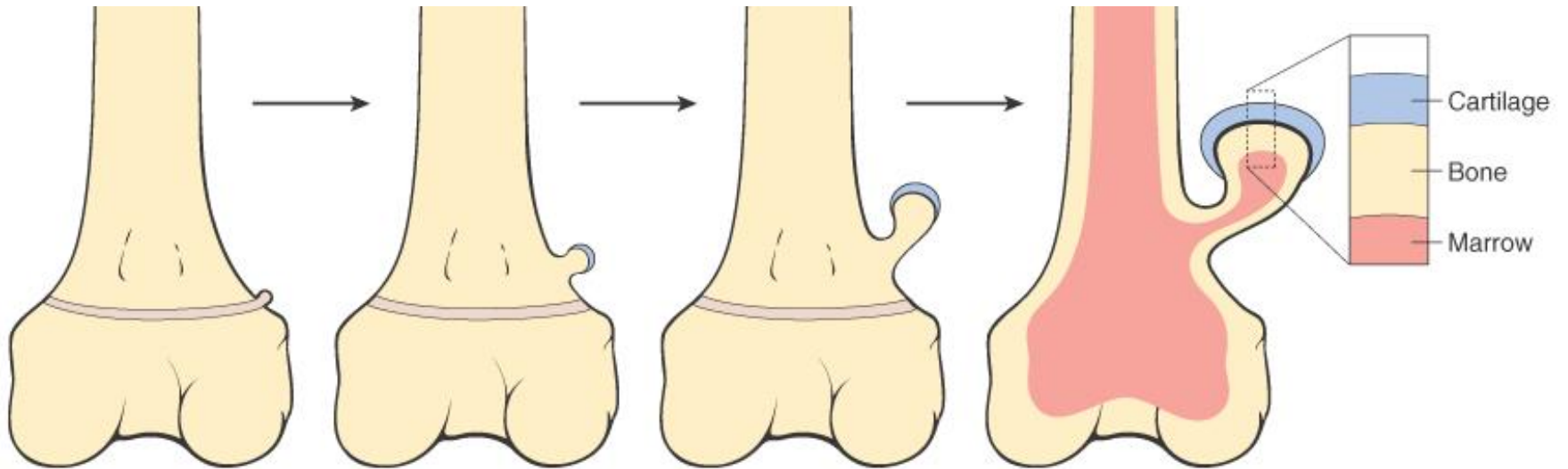
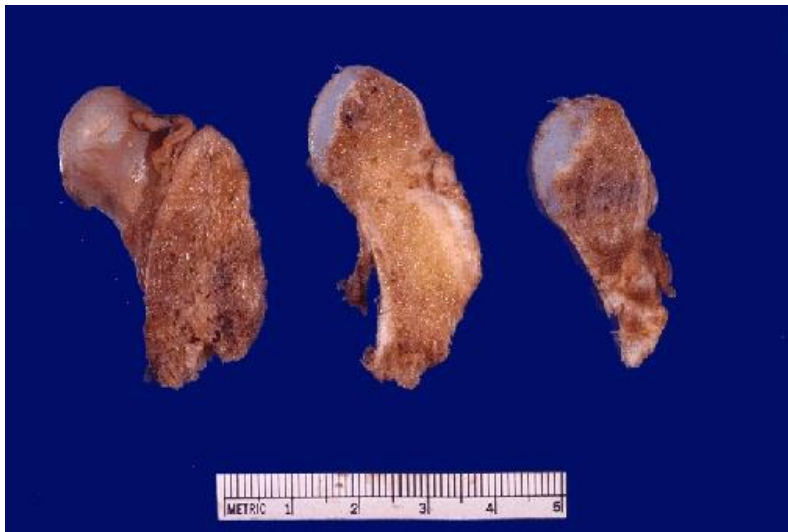
Cont..

- Most are asymptomatic but can be painful or pathological fractures may occur
- Deformities are common in multiple cases
- Growth potential is limited and most remain stable
- Incomplete resection results in recurrence
- Rare sarcomatous change is seen

Cont..

Osteochondroma (Exostosis)

- benign proliferations composed of mature bone and a cartilaginous cap
- Account for about one third of all benign tumors of the bone
- May occur in any bone; usually metaphysis of long bones (lower end of femur, upper end of humerus and upper end of tibia are most frequent)
- In children
- <1% risk of sarcomatous transformation





Cont..

Chondrosarcoma

- Second most common malignant matrix(cartilage) producing tumor
- Patients are usually ≥ 40 years of age.
- M : F = 2 : 1
- Significant cases occur in association with a pre-existing enchondroma and few develop in cases of Osteochondroma, chondroblastoma, fibrous dysplasia or Paget disease.

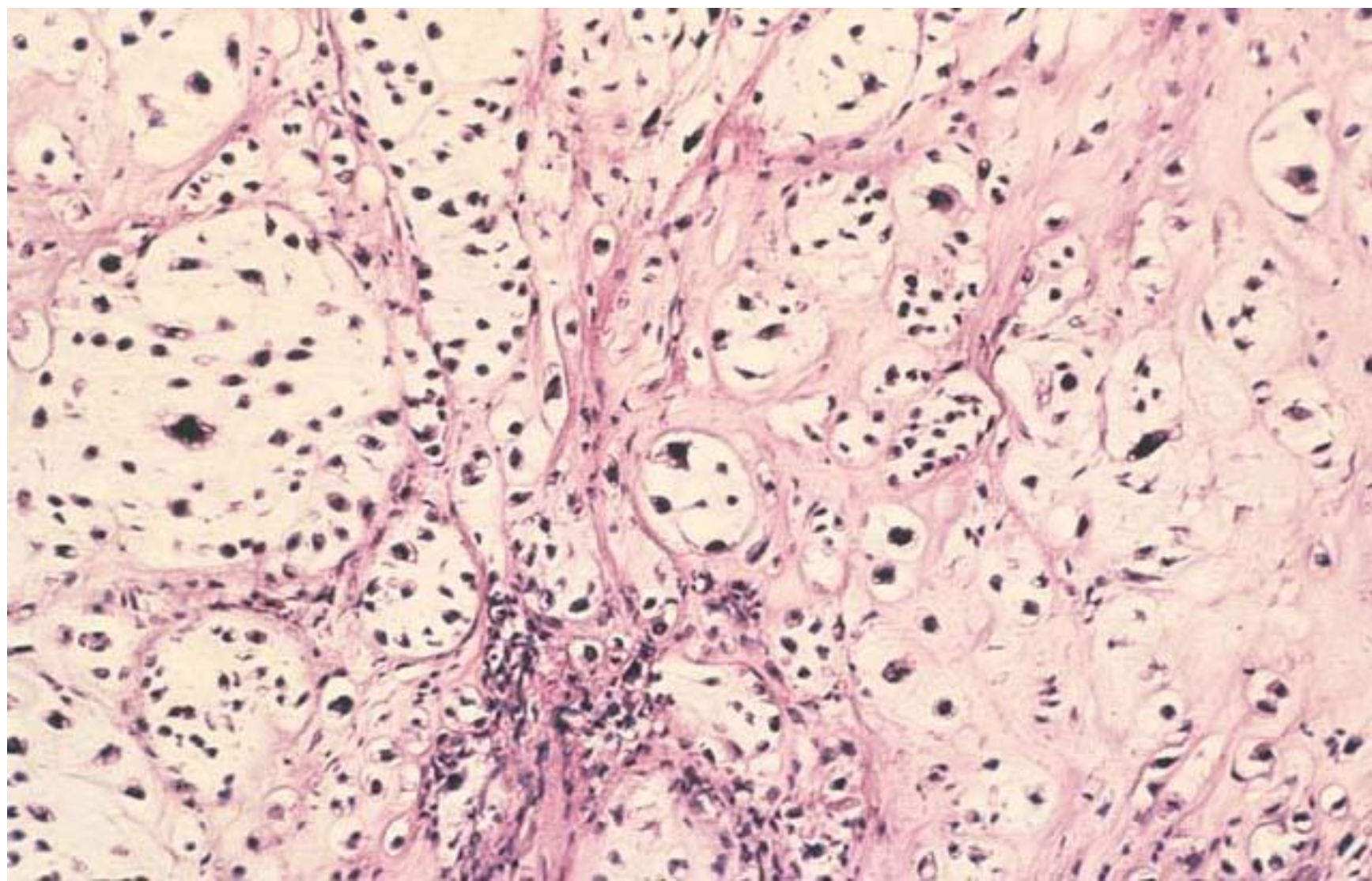
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- They arise in central portions of the skeleton; common sites of origin include the shoulder area, pelvis, proximal femur, and ribs
- Site- Intramedullary and Juxtacortical
- Microscopy-Conventional (hyaline or myxoid), Clear cell, Dedifferentiated, and Mesenchymal variants

Ill-defined margins;
fusiform thickening of
shaft; perforation of
cortex





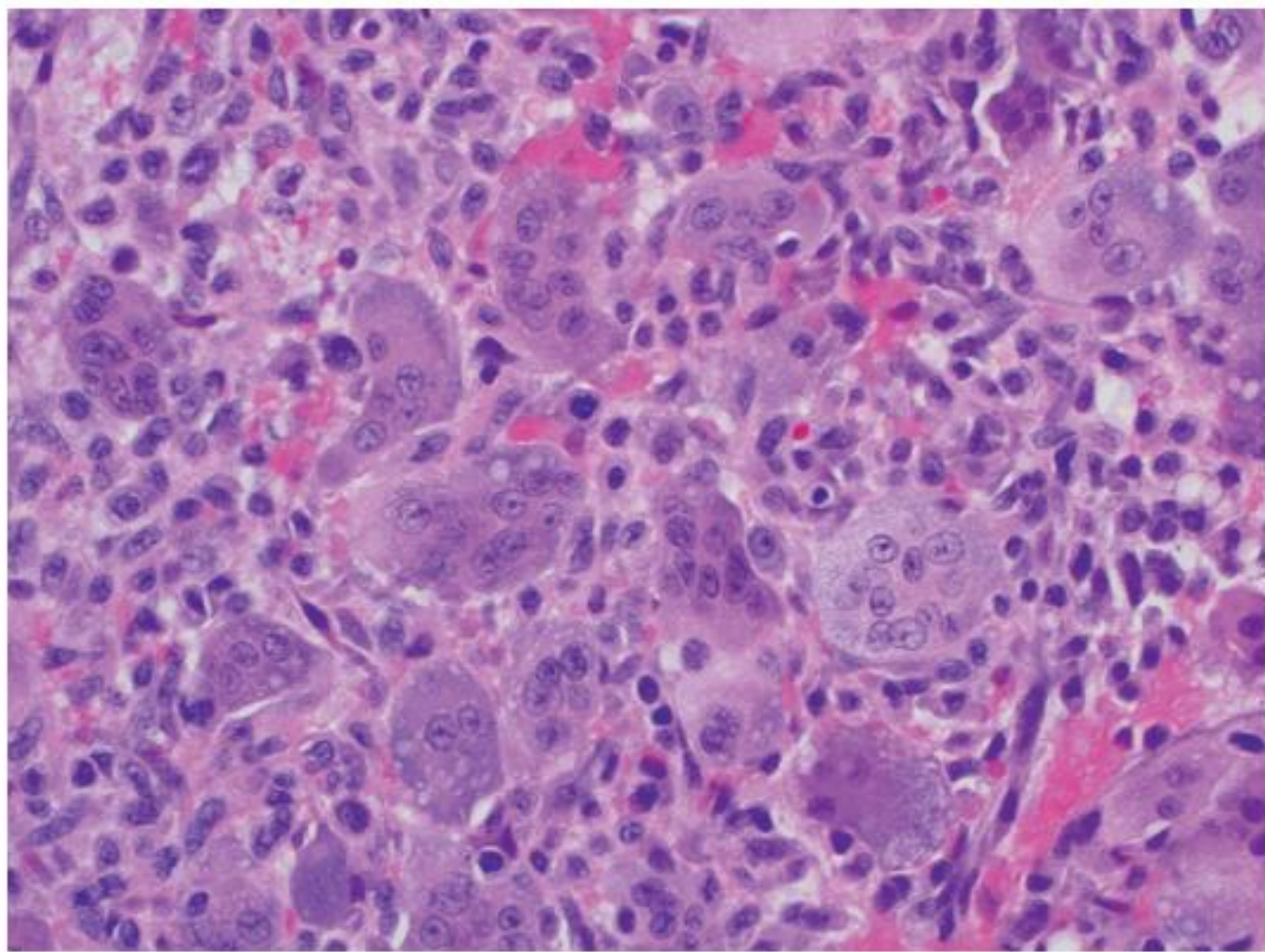


Cont..

- Microscopically, nodular pattern is seen and malignant features depend on grade of tumor
- 3 grades- depending on the cellularity , nuclear atypia and mitosis
- Prognosis depends on grade
- Patients present with painful progressively enlarging masses
- Metastases show predilection for lungs and skeleton

Giant Cell Tumor

- Giant cell tumor is so named because multinucleated osteoclast-type giant cells dominate the histology.
- It is a locally aggressive neoplasm that almost exclusively affects adults.
- Giant cell tumors arise in the epiphyses of long bones, most commonly the distal femur and proximal tibia.
- The typical location of these tumors near joints frequently causes arthritis-like symptoms.
- Occasionally, they present with pathologic fractures
- Microscopically, the tumor conspicuously lacks bone or cartilage, consisting of numerous osteoclast-type giant cells with 100 or more nuclei with uniform, oval mononuclear tumor cells in between
- Giant cell tumors are typically treated with curettage, but 40% to 60% recur locally. Up to 4% of tumors metastasize to the lungs



Cont..

Ewing's family of tumors

- Primary malignant small round cell tumors
- Differ in their degree of neural differentiation.
- Neural differentiation → PNET
- Undifferentiated → EWING SARCOMA
- 6-10% of primary malignant bone tumors.
- highly aggressive neoplasm that must be differentiated from other pediatric tumors composed of "small blue cells"

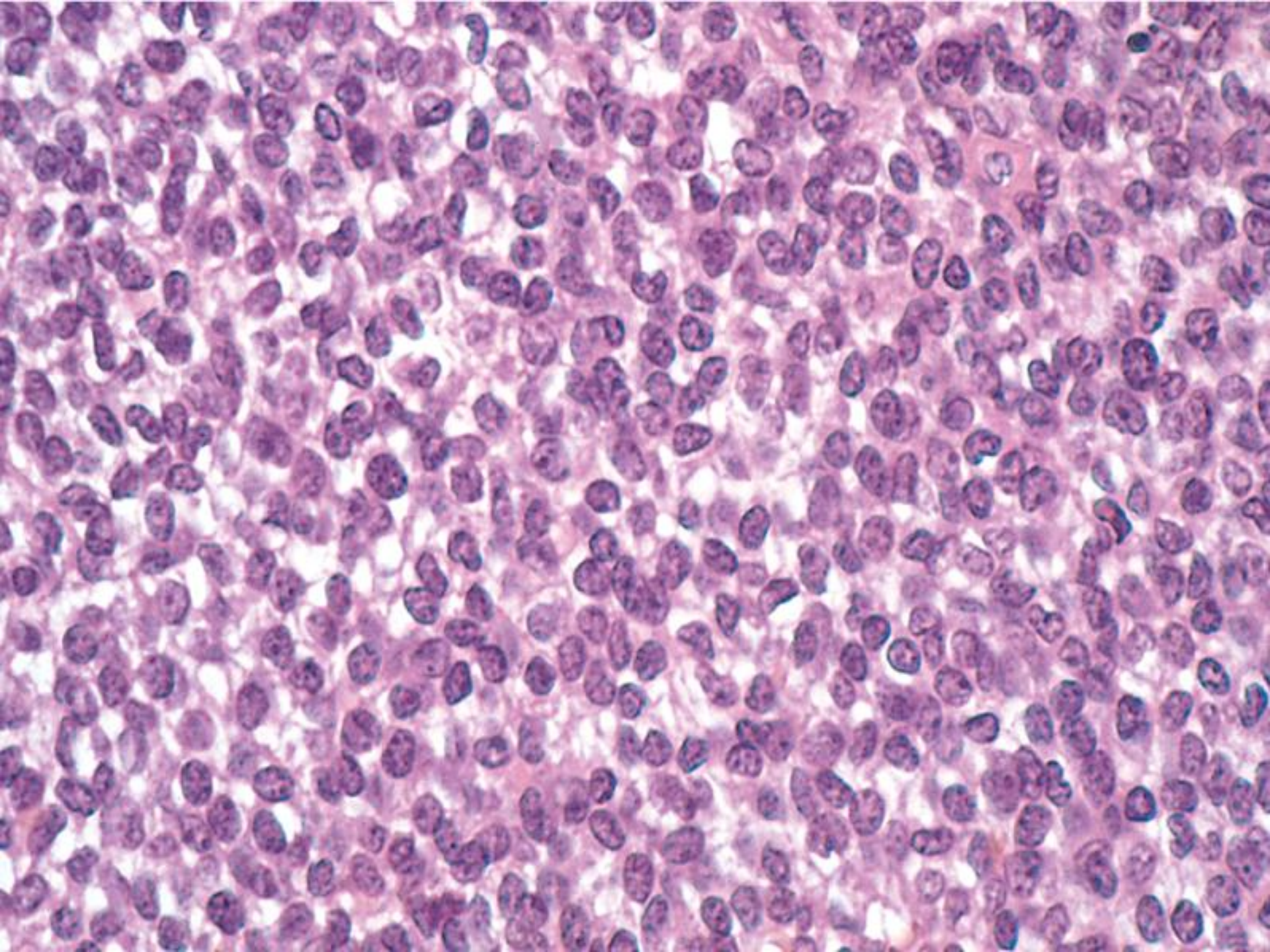
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- Ewing sarcoma is second to osteosarcoma in children
- Ewing sarcoma has the youngest age affection of malignant tumors (b/n 10 & 15yrs)
- Both arise from the medullary cavity and invade the cortex & periosteum → to produce soft tissue masses
- Tumors are tan white with areas of hemorrhage and necrosis
- The femur, tibia, and pelvis are favored sites of origin
- Usually arise in the diaphysis



Cont..

- Microscopically, sheets of small round cells that are slightly larger than lymphocytes with scanty clear cytoplasm (glycogen rich)
- Rosettes (Homer-wright rosettes) → neural differentiation
- Stroma is scanty but fibrous septa is seen
- Few mitotic figures are seen



Cont..

Clinical feature

- Painful enlarging mass the site is tender, warm and swollen.
- Systemic symptoms may be seen
- X-ray shows lytic lesions with permeation in to soft tissue
- Treatment is chemotherapy and surgical excision with or without radiotherapy
- 5yr survival reaches up to 75%

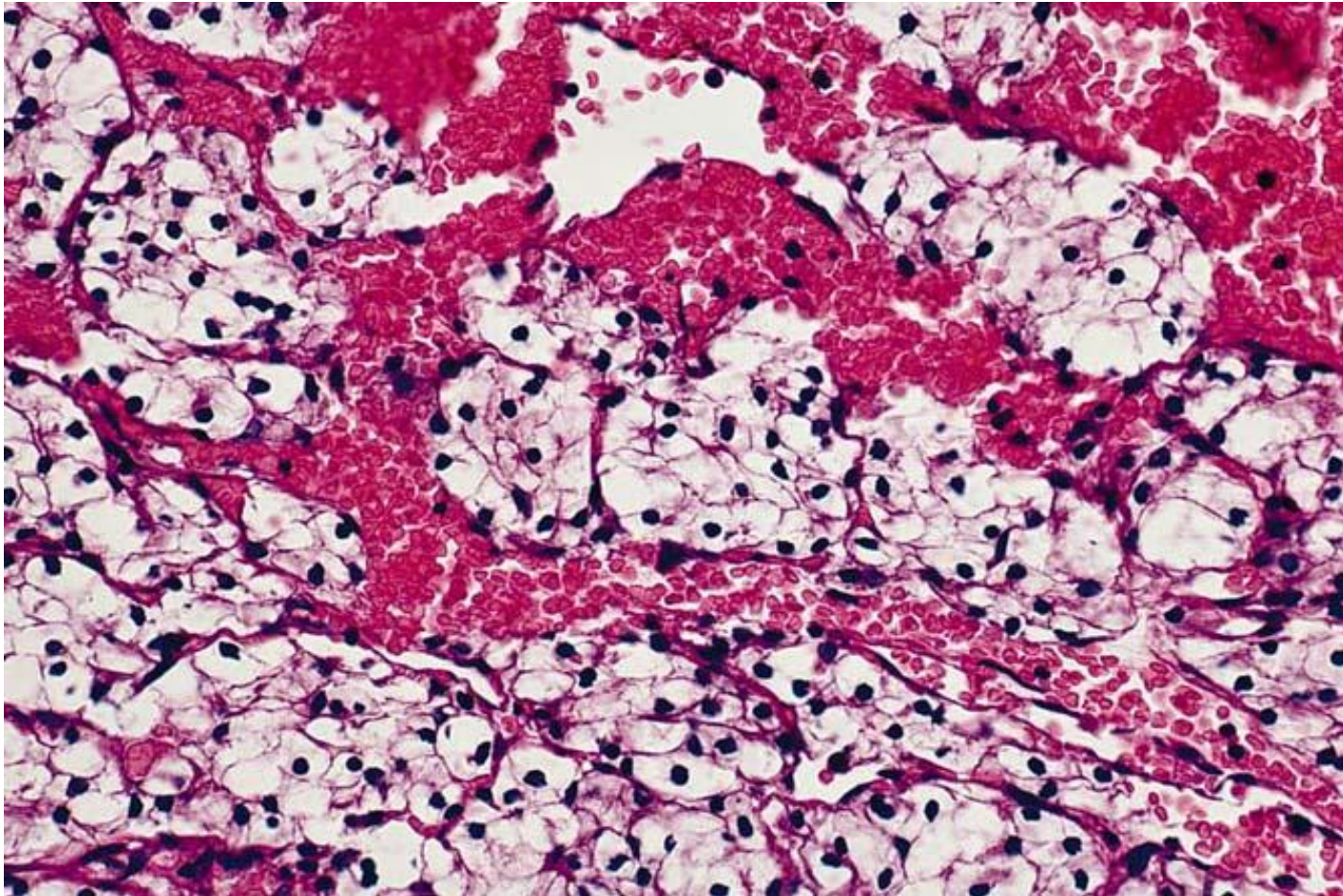
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Metastatic Disease

- Most common form of skeletal malignancy
- Direct extension or Lymphohematogeneous dissemination
- 75% of cases originate from cancers of the **prostate, breast, kidney and lung**
- In children- neuroblastoma, Wilms tumor, osteosarcoma, Ewing sarcoma and rhabdomyosarcoma
- Typically Multifocal
- Thyroid and kidney carcinomas -solitary

Cont..

- Most are in the axial skeleton (vertebral column, pelvis, ribs, skull & sternum) – marrow
- Metastasis to small bones of hands and feet is uncommon
- X-ray – lytic, blastic or mixed
 - Kidney, lung, GI, and melanoma – lytic lesions
 - Prostatic adenocarcinoma – blastic lesions



Renal cell carcinoma of clear cell type metastatic to bone. Notice the marked fresh hemorrhage that is a characteristic feature of this tumor.



Ill-defined lytic lesion in midshaft of fibula produced by metastasis of lung carcinoma

Diseases of the Joints

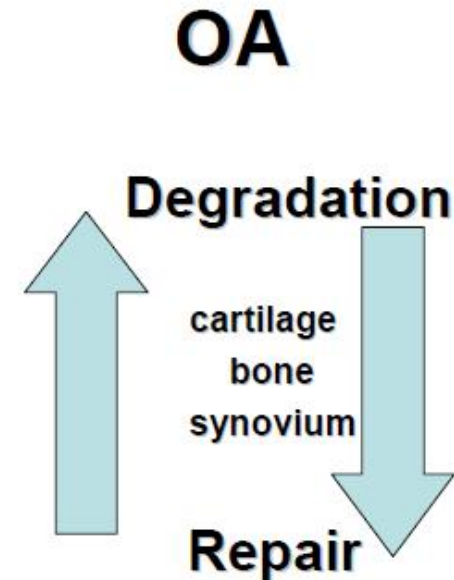
Osteoarthritis (*degenerative joint disease*)

- Is the most common disorder of the joints
- Fundamental feature -degeneration of the articular cartilage
- May arise without any obvious predisposing factors as an aging process (*primary or idiopathic*)
- *Secondary osteoarthritis-usually* in young in a previously deformed joints or in some metabolic disorders

Cont..

Pathogenesis

- Normally balanced articular cartilage degradation and replacement
- In osteoarthritis, this process is disturbed by a variety of influences



Cont..

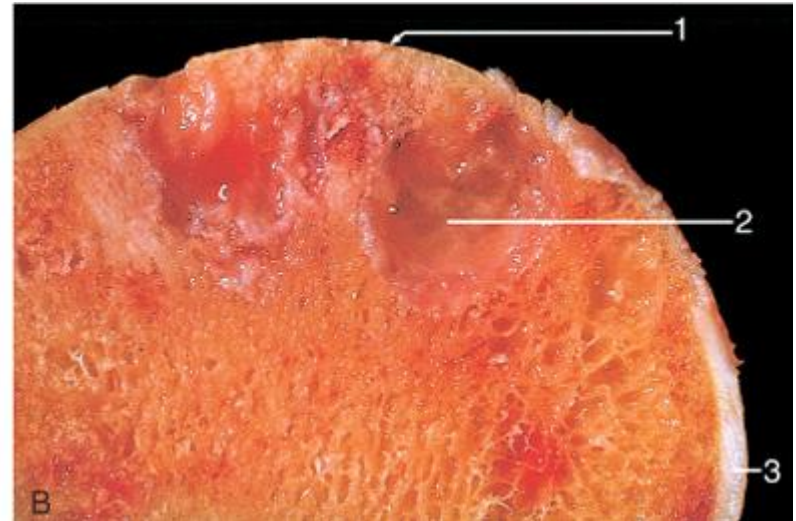
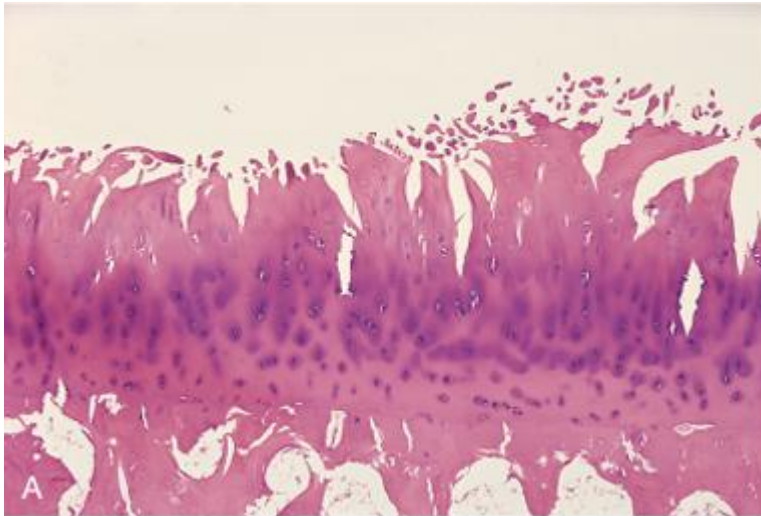
- The most important of these influences are *aging and mechanical effects*
- Characterized by significant changes in both the composition and the mechanical properties of cartilage
- Early changes include , increased water and decreased proteoglycans
- Attempts to repair by deep chondrocytes

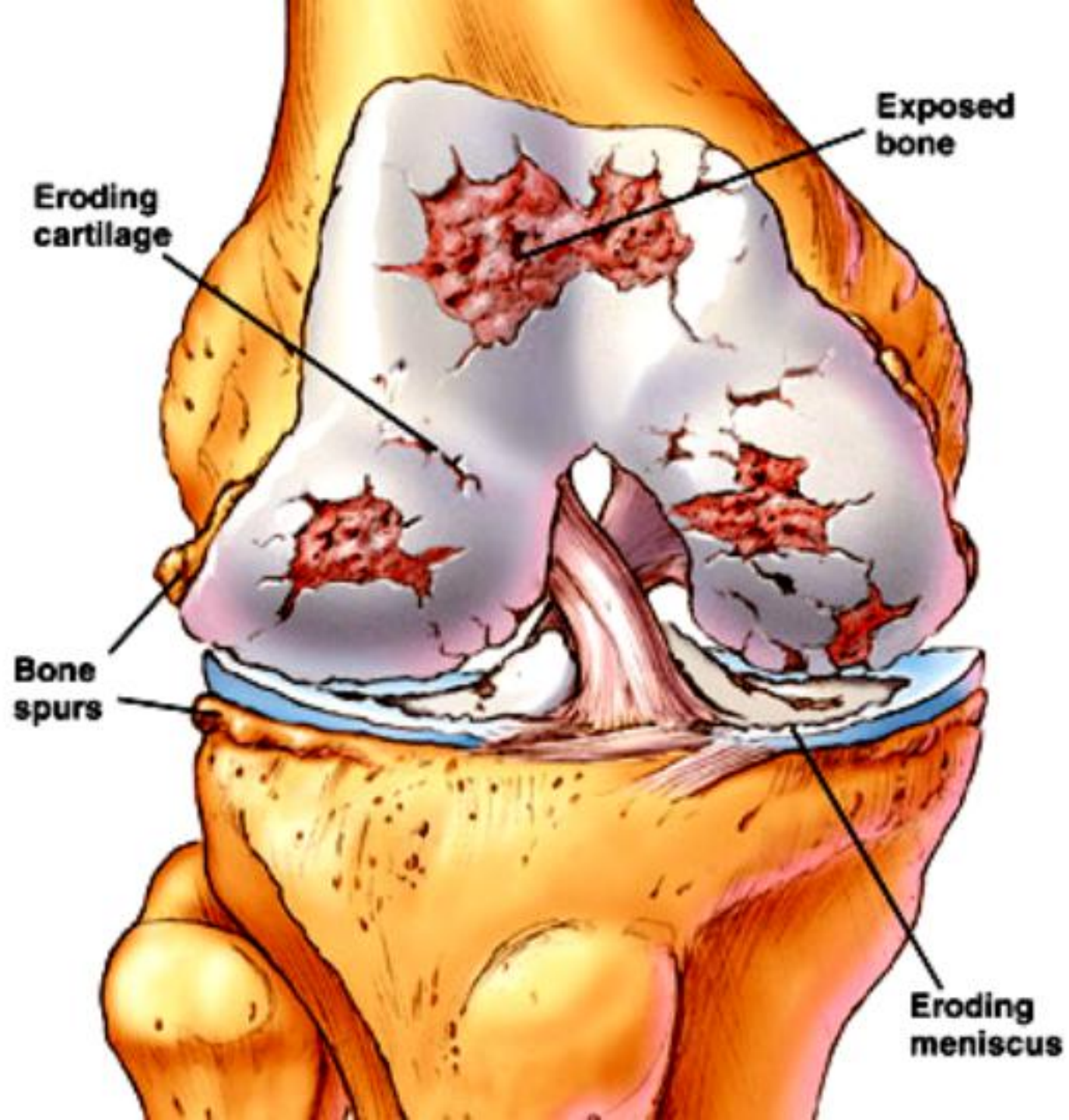
Cont..

Morphology

- Fibrillation (splitting) at the articular surface
- Erosion of the articular cartilage
- Thickened subchondral bone
- Fragments of cartilage and bone are often dislodged to form free-floating "joint mice"
- Bone proliferation occurs at the margins of the joints to produce bony excrescences, termed **osteophytes**

Cont..







Heberden
nodes- in
females,not in
males
Prominent
osteophytes

Cont..

Clinical features

- Gradual onset of symptoms
- the hips, knees, lower lumbar and cervical vertebrae, proximal and distal interphalangeal joints of the fingers commonly involved
- Asymptomatic or common complaints include joint stiffness and deep, aching pain, particularly in the morning which worsens with use.
- Some degree of joint swelling is common, and small effusions may develop

Cont..

Rheumatoid arthritis

- A systemic, chronic, inflammatory disorder characterized by **progressive arthritis , production of rheumatoid factor, and extra-articular manifestations**
- Nonsuppurative proliferative and inflammatory synovitis that often progresses to destruction of the articular cartilage and ankylosis of the joints

Cont..

- Unknown cause
- Autoimmunity plays a role in chronicity and progression
- 1% Of world population affected
- F:M=3:1
- In 40-70yrs

Cont..

Pathogenesis

- An autoimmune diseases triggered by exposure of a genetically susceptible host to an unknown arthritogenic antigen

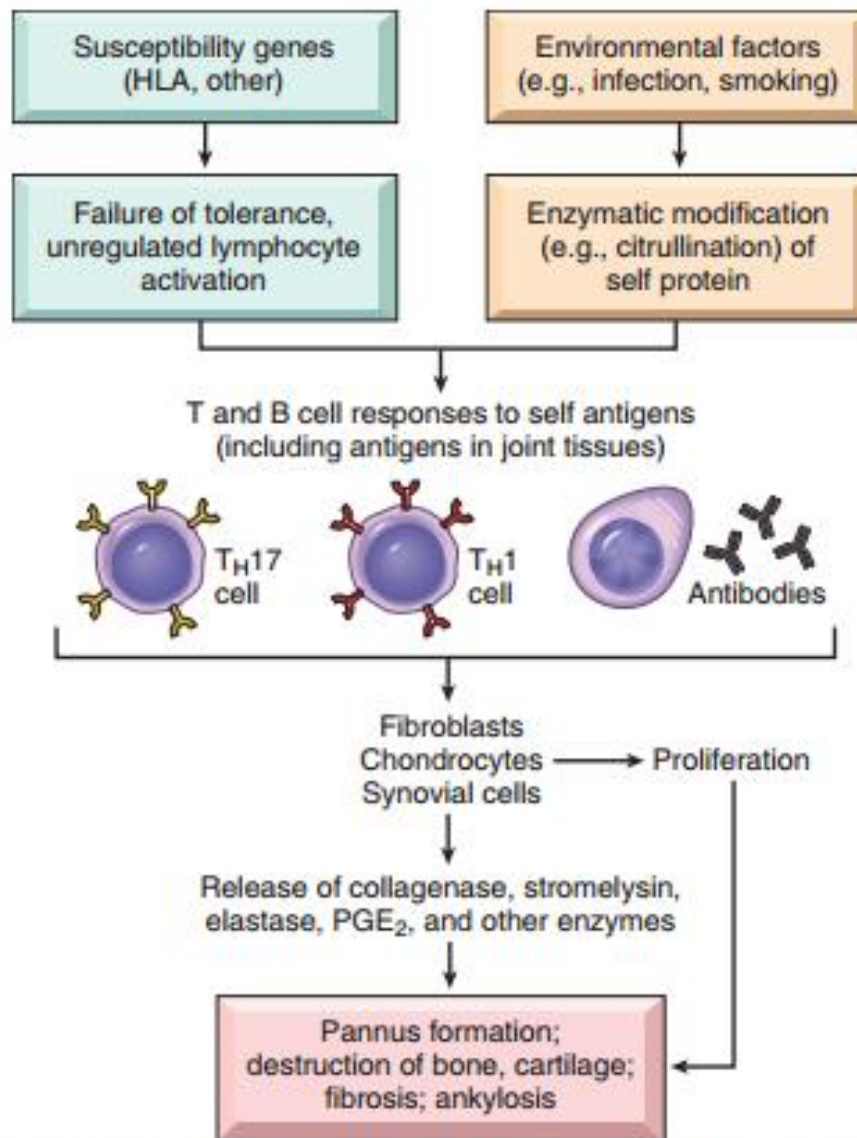


Figure 20–18 Major processes involved in pathogenesis of rheumatoid arthritis.

Cont..

Morphology

Joints

- Perivascular mononuclear inflammation of the synovial stroma , edema and hyperplasia...bulbous fronds
- Granulation tissue
- 'Rice bodies' and organization of synovial fibrin
- Neutrophils in the synovial fluid but not deep in the stroma



Cont..

- Juxta-articular erosions , Subchondral cysts, and osteoporosis
- Pannus formation....fibrous ankylosis.....ossification and bony ankylosis
- Pannus-a mass of synovium and synovial stroma consisting of inflammatory cells , granulation tissue,& fibroblasts which grows over articular cartilage and causes erosion

Cont..

- Skin-rheumatoid nodules(25%)....elbows ,occiput and lumbosacral area
- Blood vessels- obliterative endarteritis.....peripheral neuropathy ,ulcers and gangrene

Cont..

Clinical course

- Variable
- Insidious onset
- 10% acute onset with severe symptoms
- Generally small joints are involved before the larger ones the hip joints involved late, if at all
- The lumbosacral region is typically spared

Cont..

- Fusiform hot swollen joints, morning stiffness improves with activity
- Symmetrical involvement of joints
- Radial deviation of the wrist and ulnar deviation of the fingers
- 'swan neck' deformity
- Baker cysts : synovial cyst in the popliteal fossa

'Swan neck' deformity



Cont..

- Elevated ESR and hypergammaglobulinemia
- Rheumatoid factor positive in 80% of RA
- Felty syndrome-RA+ neutropenia+ splenomegaly
- Caplan syndrome.....in pneumoconiosis
- Amyloidosis

Cont..

Juvenile rheumatoid arthritis (JRA)

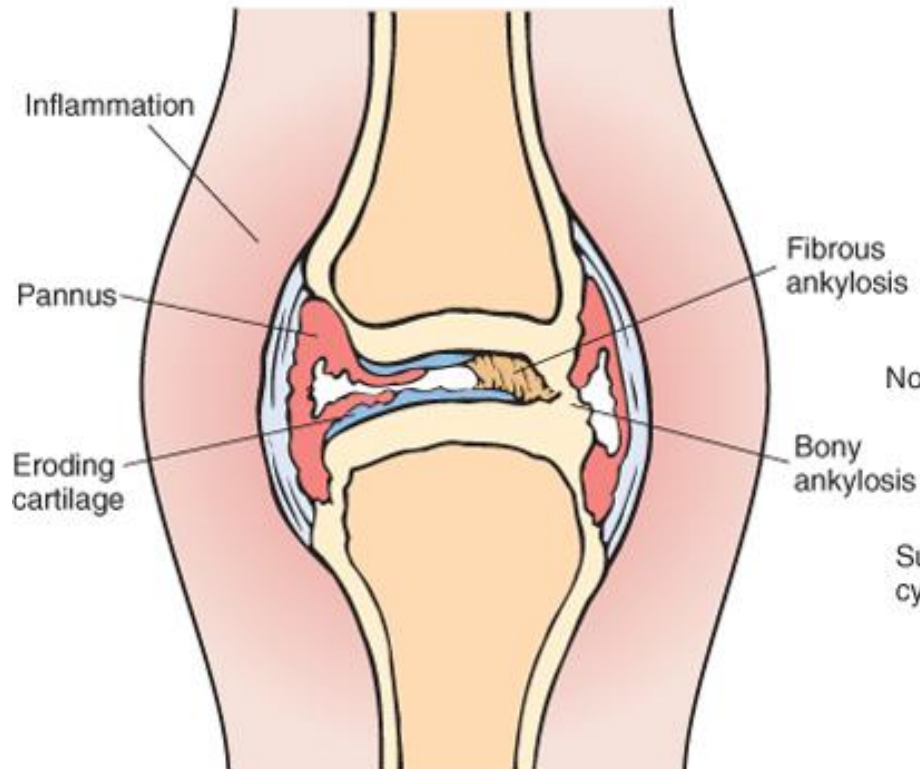
- Occurs in children younger than 16 years of age
- More common in girls
- RF is usually absent
- Oligoarthritis is more common
- Frequent systemic onset
- often large joints are affected
- Antinuclear positivity is common

Cont..

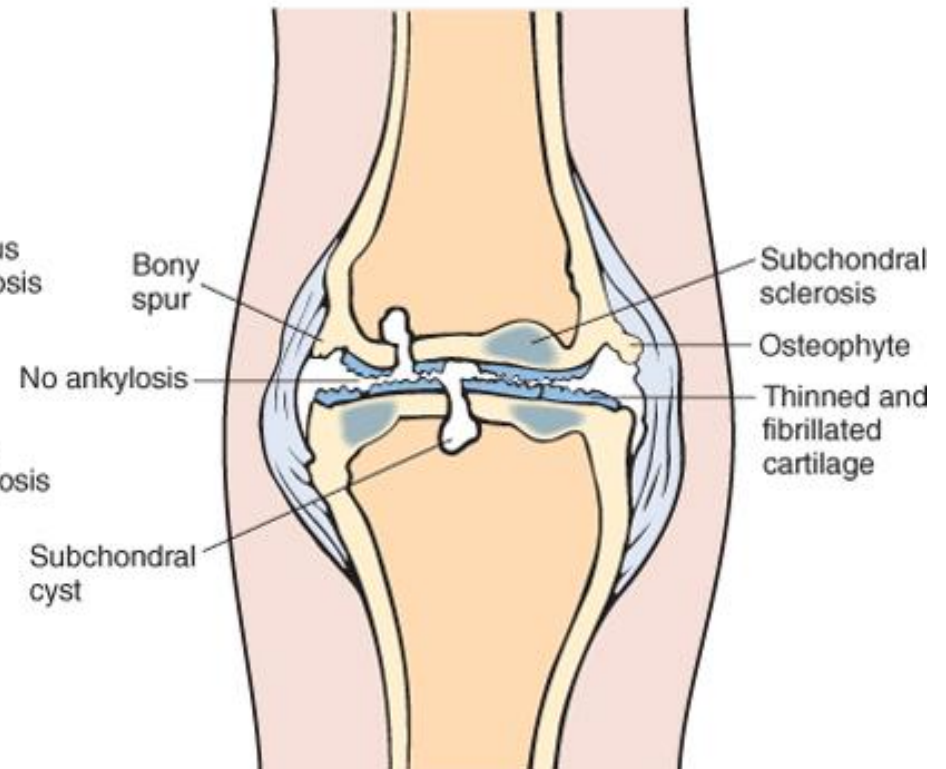
- Polyarticular JRA
 - Disabling arthritis predominates
 - > 5 joints involved
- Pauciarticular JRA
 - Arthritis limited to a few joints
 - Uveitis with the potential for blindness



RHEUMATOID ARTHRITIS



OSTEOARTHRITIS



Cont..

- Infectious arthritis

Suppurative arthritis

- Routes of infection

Hematogenous route

- Most common

- Seeding of joint during bacteremia

Spread from adjacent site of infection

Direct inoculation

Cont..

- Gonococci, staph. aureus, streptococci, H.influenza, G-negative bacilli
- Usually monoarticular , tender , painful , swollen, erythematous large joints
- Joint aspiration- cloudy ,clots easily , high neutrophil count , positive G-stain and culture in 50-70% of cases

Cont..

Tuberculous arthritis

- Chronic progressive monoarticular disease
- All age especially in adults
- Hematogenous or from adjoining TB osteomyelitis
- Systemic symptoms may or may not be present

Cont..

- Confluent granulomas and Pannus
- Severe destruction with fibrous ankylosis and obliteration of the joint space
- Hips ,knees and ankles commonly affected

Gout and gouty arthritis

- group of diseases characterized by an increased serum uric acid level and deposition of monosodium urate crystals in joints ,soft tissue around joints and skin

Cont..

- Gout is the common end point of a group of disorders that produce hyperuricemia
- Leads to- acute arthritis ,Chronic gouty arthritis and tophi
- Most , but not all patients with gout also develop urate nephropathy
- Hyperuricemia is a sine qua non , but not a sole determinant

Cont..

- Primary(90%)-Unknown cause ,gout is the initial manifestation
 - overproduction or decreased excretion
- Secondary(10%)-known cause ,gout is not the main clinical dysfunction
 - over production e.g in leukemias or underexcretion e.g renal disease

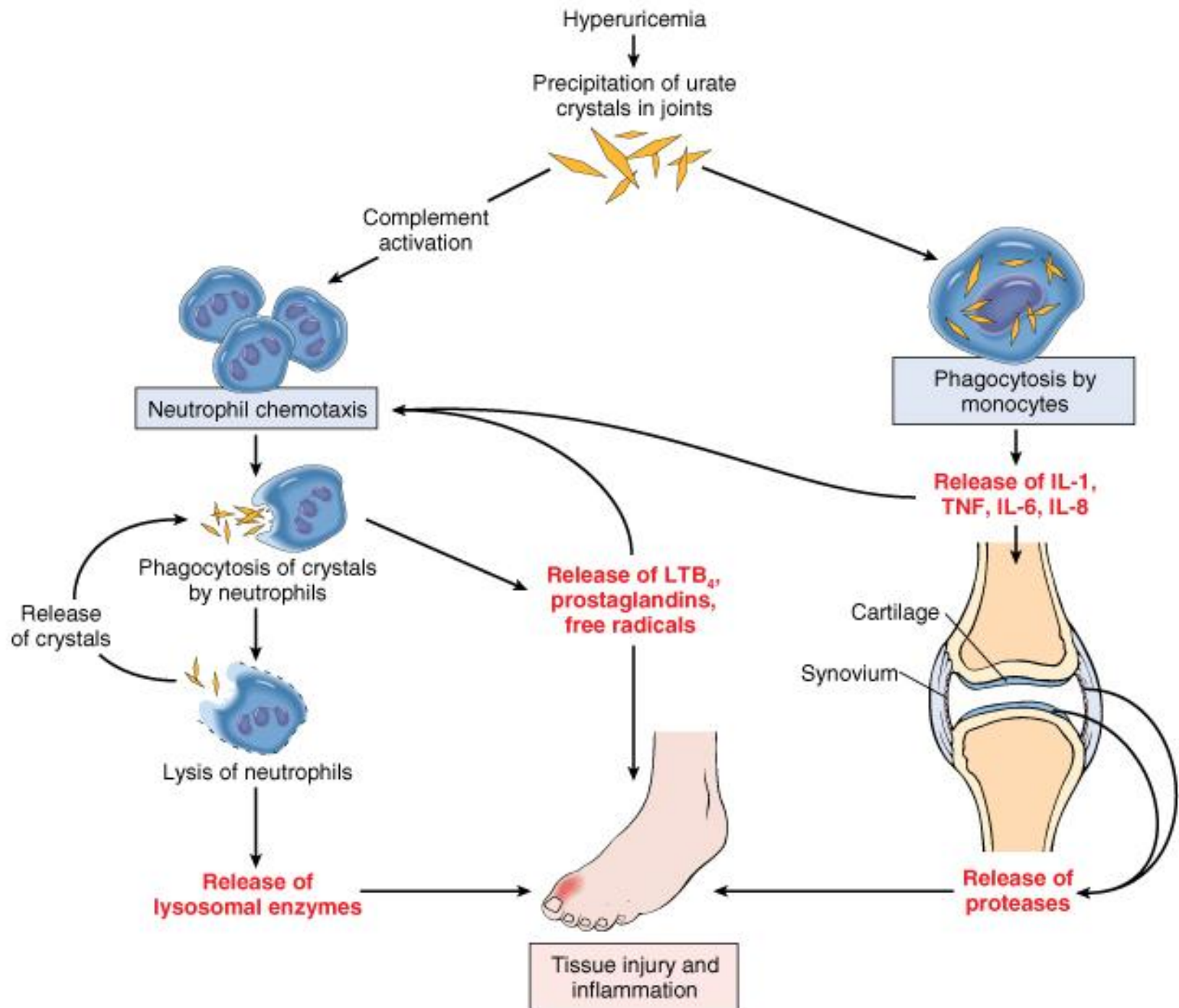
Cont..

Factors which favor the development of arthritis

- Age and duration of hyperuricemia
- Genetic predisposition
- Heavy alcohol consumption
- Obesity
- Drugs e.g thiazides
- Lead toxicity

Cont..

- Central to the pathogenesis of the arthritis is precipitation of monosodium urate crystals into the joints



Pathogenesis of hyperuricemia

- uric acid is the end product of the catabolism of purines derived either from diet or synthesized denovo
- uric acid is eliminated from the body mostly through urine
- values of $>7.0\text{mg/dl}$ exceeds blood saturation at body temperature
- Hyperuricemia can result from overproduction of uric acid, decreased urinary excretion or a combination of both

Cont..

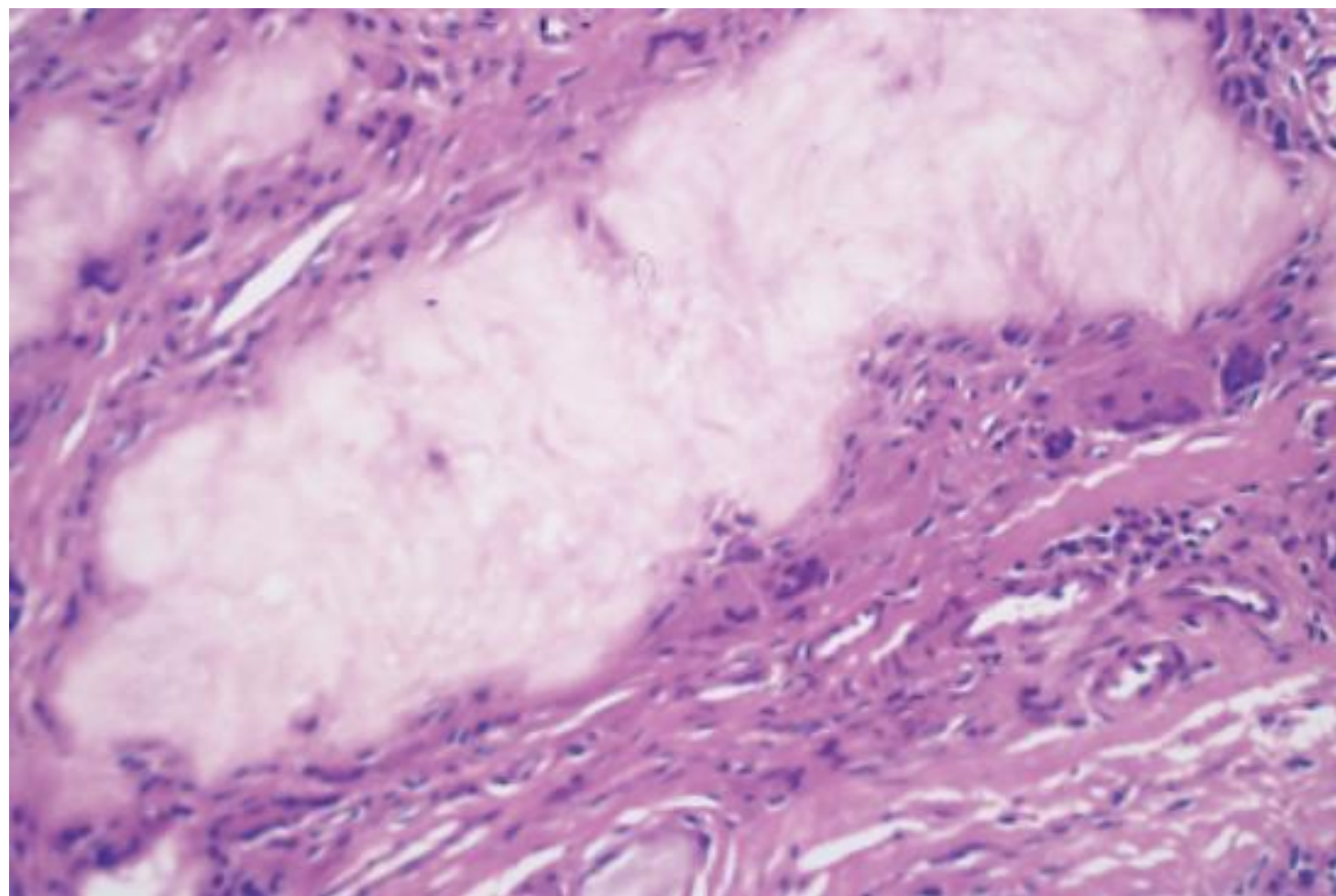
Morphology

- Acute arthritis-acute inflammation against needle shape crystals
- Chronic Tophaceous arthritis- repetitive episodes ending in chronic inflammation, Pannus formation and complicated by fibrous and bony ankylosis
- Tophi are hallmark of gout
- Gouty nephropathy- interstitial or renal uric acid stones and obstruction

Cont..

Tophi

- Extracellular deposits of these crystals (tophi) are surrounded by foreign body giant cells
- macroscopically it appears as chalky which deposits on the surfaces of extraarticular structures and soft tissues
- Classic locations are ear, head, olecranon bursa and in the Achilles tendon



Tophi



Cont..

Clinical Features

Four steps in the clinical course

1. Asymptomatic hyperuricemia
 - precedes clinically evident gout by many yrs
2. Acute gouty arthritis
 - Initially monoarticular involvement later polyarticular with fever
3. Intercritical gout
 - Asymptomatic interval b/n attacks
4. Chronic tophaceous gout

tumors & tumor-like lesions of the joints

Ganglion cyst

- 1-1.5cm cyst almost always located near a joint capsule or tendon sheath
- Common in the wrist joint area
- Firm , fluctuant, pea-sized translucent nodule
- As a result of cystic or myxoid degeneration of connective tissue
- Cyst wall with no lining epithelium, no communication with the joint space

Cont..

Synovial cyst

- Herniation of synovium through a joint capsule or massive enlargement of bursa
- E.g baker cyst
- Synovial lining may be hyperplastic and contain inflammatory cells and fibrin

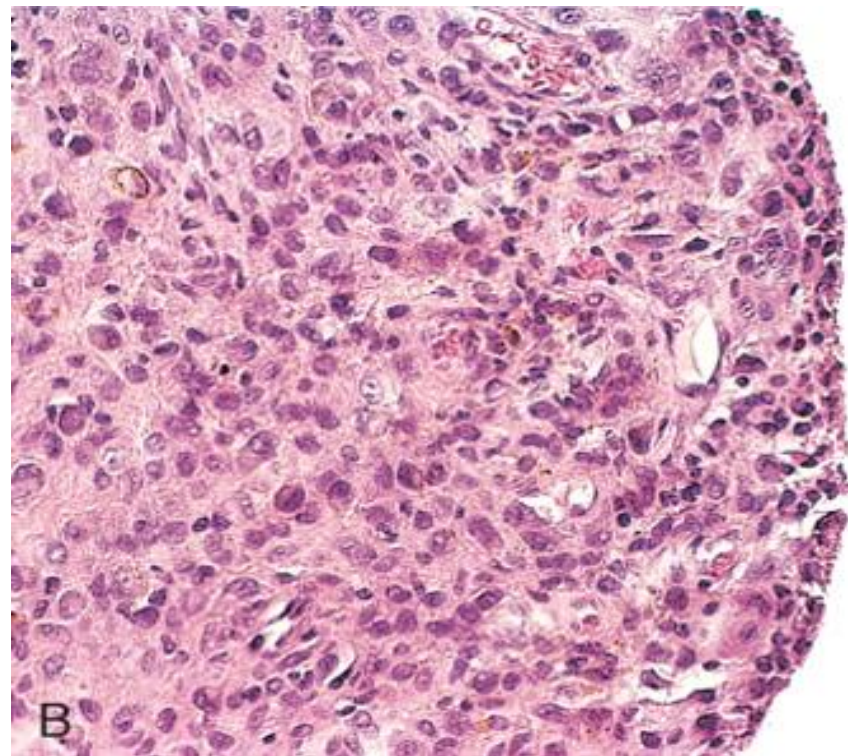
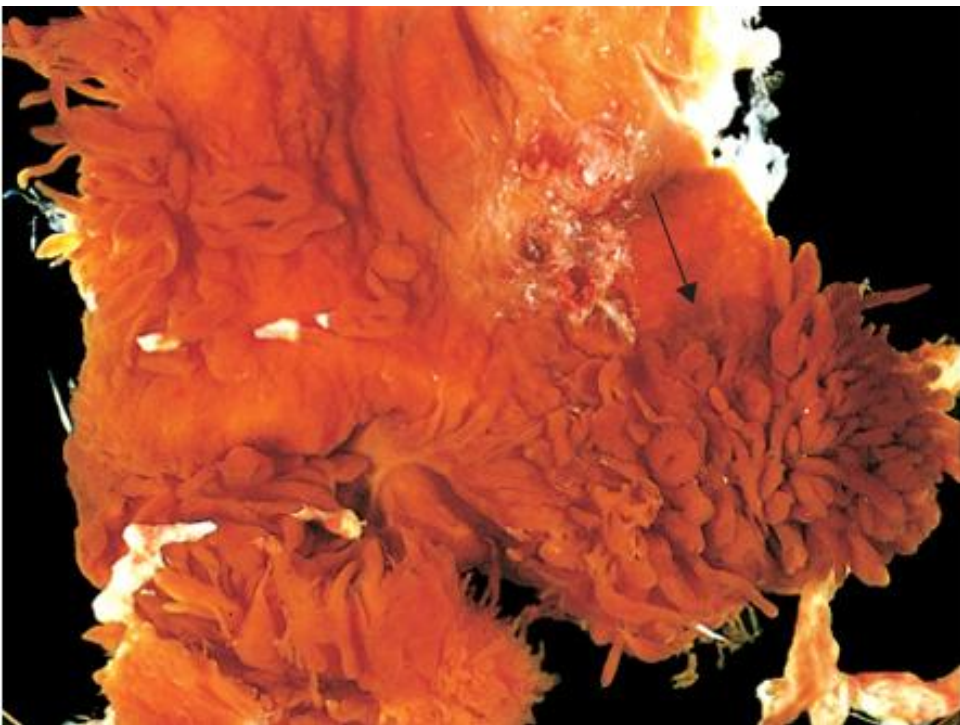
Cont..

Pigmented villonodular synovitis and giant cell tumor of the tendon sheath

- Several closely related benign neoplasms in the synovial lining of joints, tendon sheaths and bursae
- PVNS-involves synovium of a joint
- GCT of the tendon sheath-localized nodular tenosynovitis
- Both in the 20s and 40s
- PVNS involves one or more joints

Cont..

- Morphology
- Gross-both red brown to mottled orange-yellow
- The smooth synovium is converted in to finger-like projections and nodules
- GCT-localized well circumscribed
- Microscopy- synoviocyte-like cells , infiltrate the subsynovial compartment
- Hemosiderin deposits , foamy macrophages, multinucleated giant cells and zones of sclerosis



Soft tissue tumors

- Mesenchymal proliferations in the extraskeletal , non-epithelial tissues of the body which recapitulate muscle , fat, fibrous tissue, vessels and nerves
- **Benign, malignant and intermediate** (low-grade malignant – locally aggressive, can recur, no metastatic potential)

Cont..

- Originate from **primitive mesenchymal stem cells**
- **Classification** according to their **differentiation lines**
(e.g. liposarcoma is not a tumor arising from lipoblast but exhibiting lipoblastic differentiation)

Classification of soft tissue tumors

Lipomatous tumors

Lipoma

Liposarcoma

Smooth muscle tumors

Leiomyoma

Leiomyosarcoma

Skeletal muscle tumors

Rhabdomyoma

Rhabdomyosarcoma

Fibroblastic tumors

Nodular fasciitis

Fibromatoses

Fibrosarcoma

Fibrohistiocytic tumors

Benign fibrous histiocyoma

Malignant fibrous histiocyoma

Vascular tumors

Hemangioma

Angiosarcoma

Tumors of peripheral nerves

Schwannoma

Neurofibroma

Malignant peripheral nerve sheath tumor

Tumors of uncertain origin

Synovial sarcoma

Lipoma

- The most common benign soft tissue tumor of adulthood
- Subcutaneous tissue of the trunk and limbs in the middle-aged and elderly
- Soft, slowly growing mass
- Microscopy-well-defined lobules of mature adipose tissue

Cont..

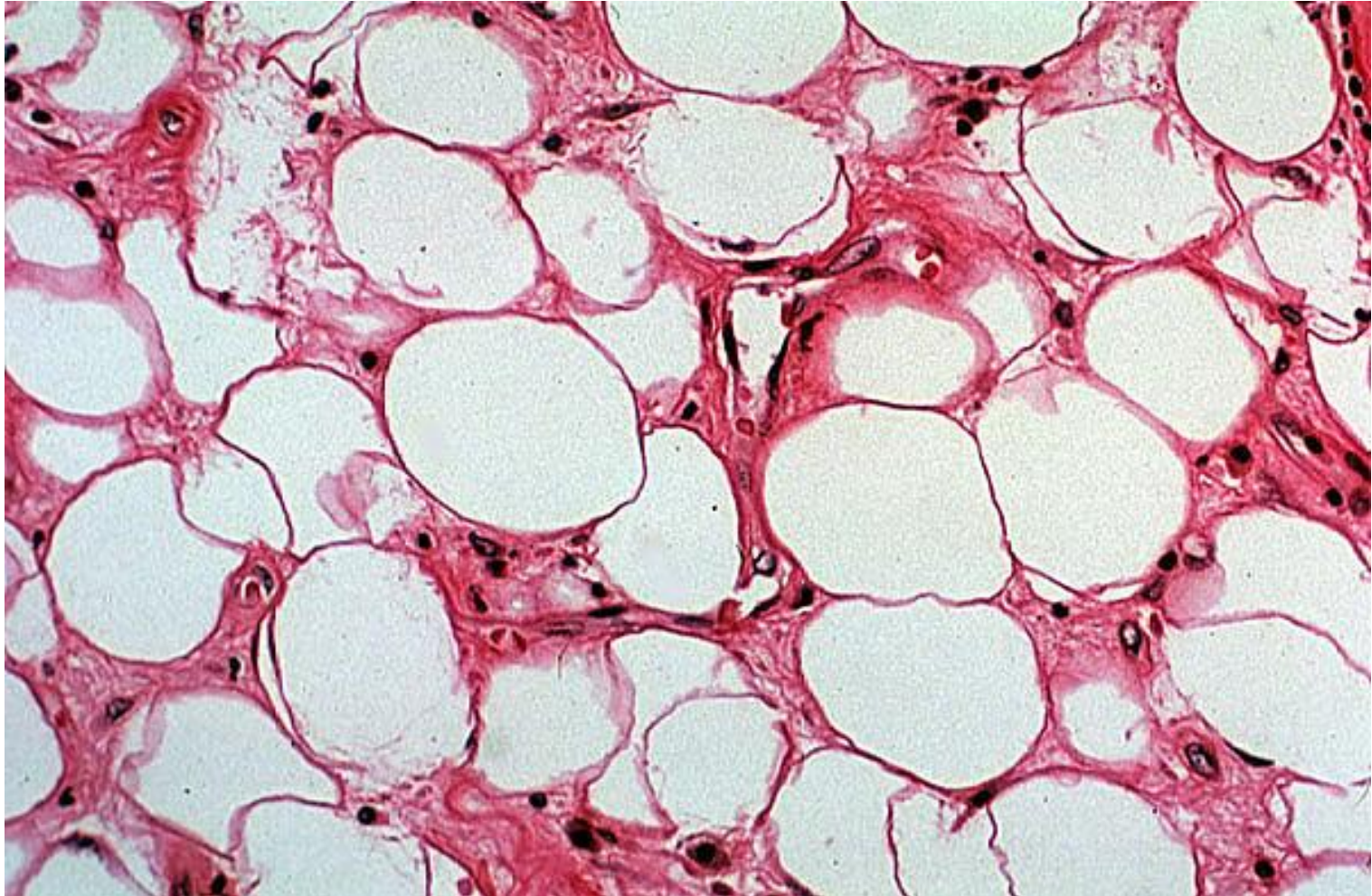
variants:

- **Angiolipoma:** thin-walled small blood vessels occupy significant portion of the lesion
- **Spindle-cell lipoma:** mixture of mature adipocytes, short bundles of collagen and small uniform spindle cells



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Cont..



Cont..

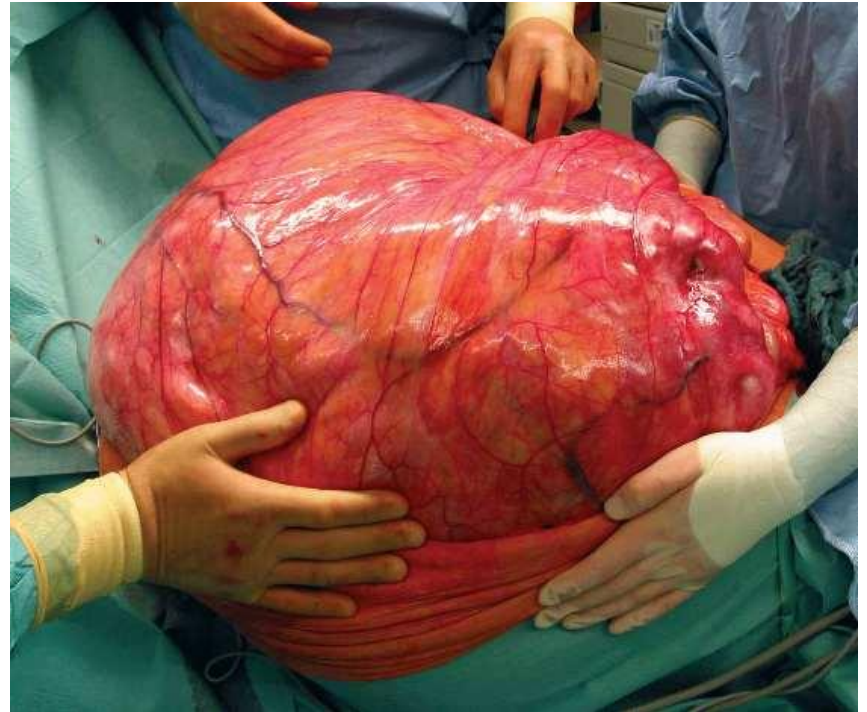
Liposarcoma

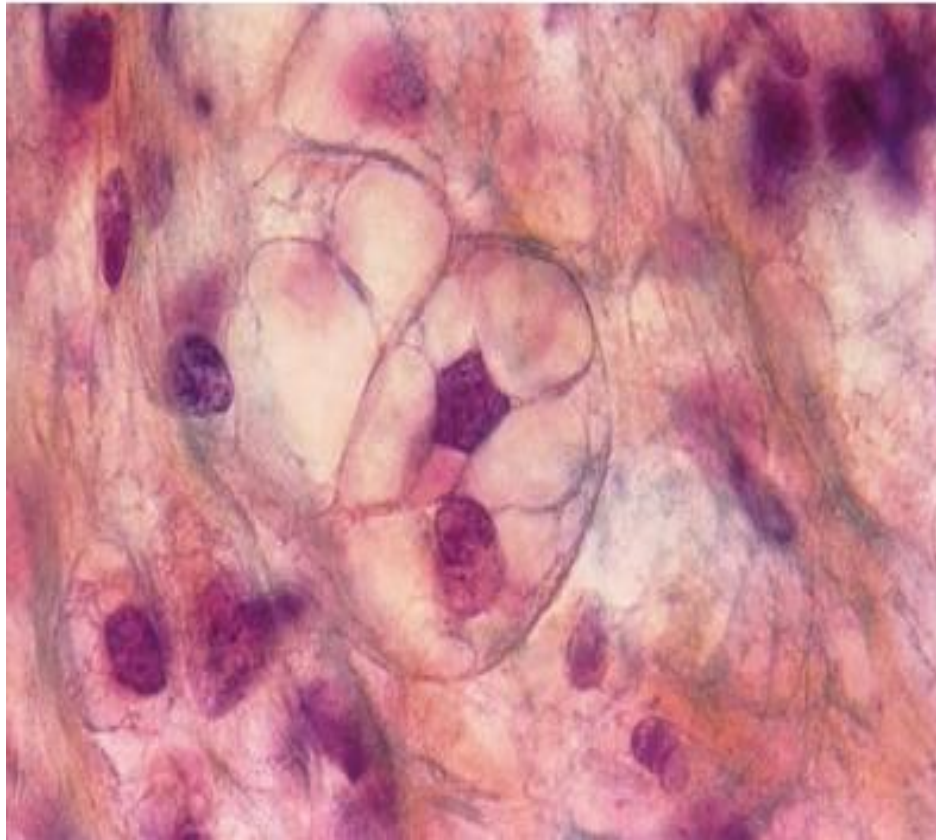
- The most common malignant soft tissue tumor
- Adults (peak incidence 40-60 years)
- lower limb and retroperitoneal space
- multivacuolated **lipoblast** and **chicken wire vessels** are key diagnostic histologic features feature

Subtypes:

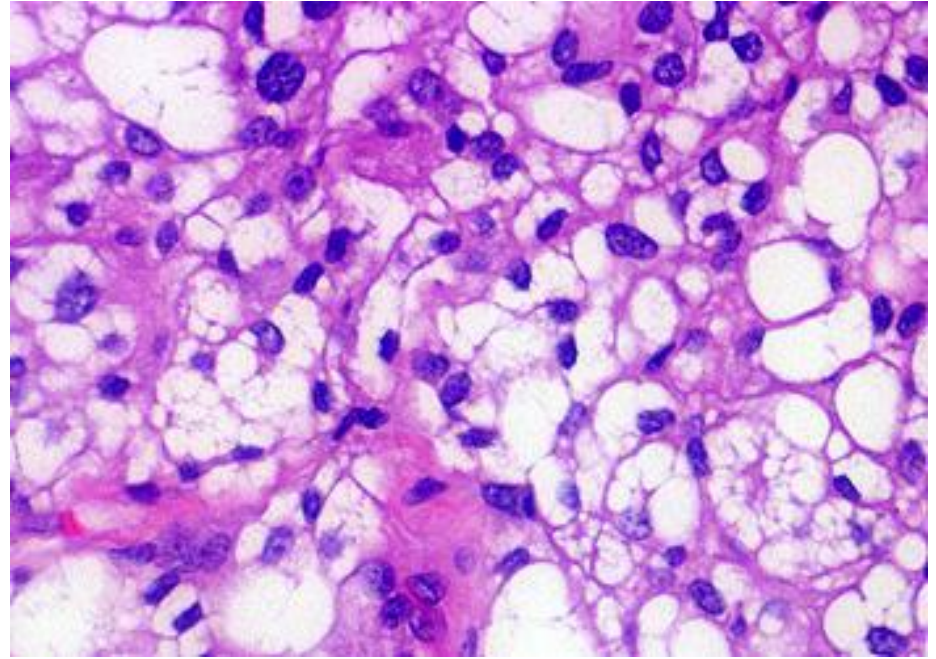
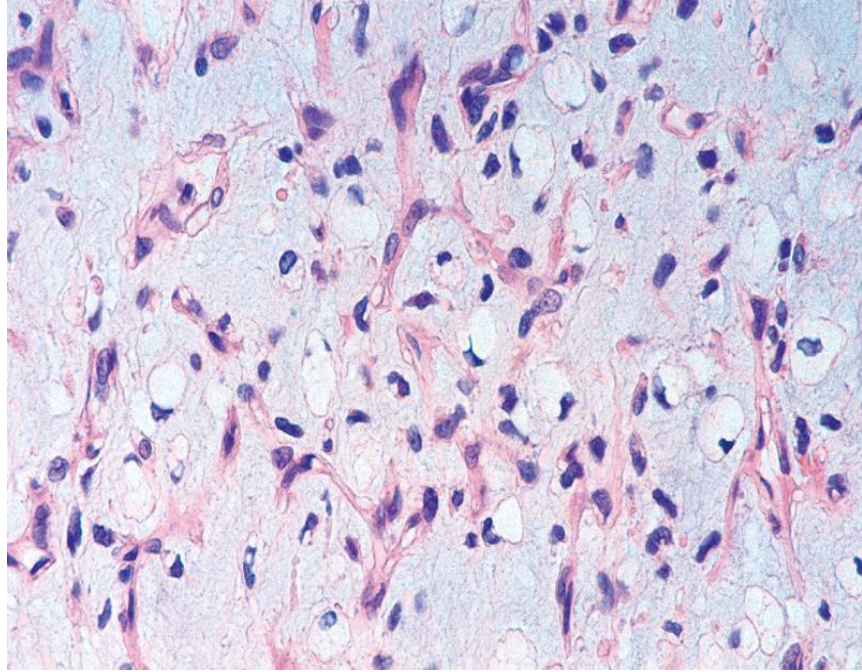
- Well-differentiated, myxoid, round cell, pleomorphic liposarcoma with decreasing prognosis and increased risk of recurrence and metastasis

Cont..





lipoblast. The hyperchromatic nuclei of lipoblasts are scalloped,



Cont..

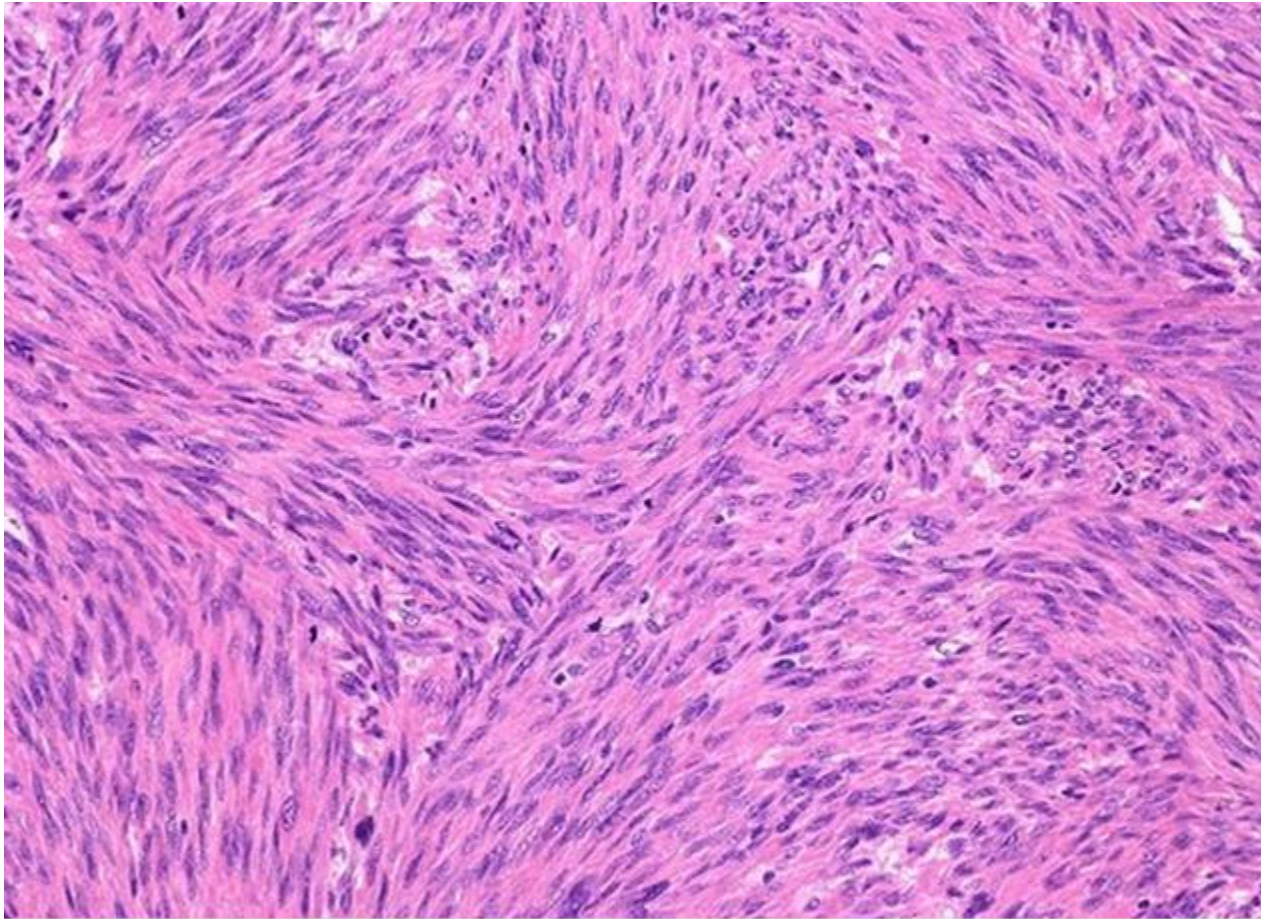
Leiomyoma

- Skin, subcutaneous tissue, uterus, gastrointestinal tract
- Microscopy-interlacing bundles of well-differentiated smooth muscle cells

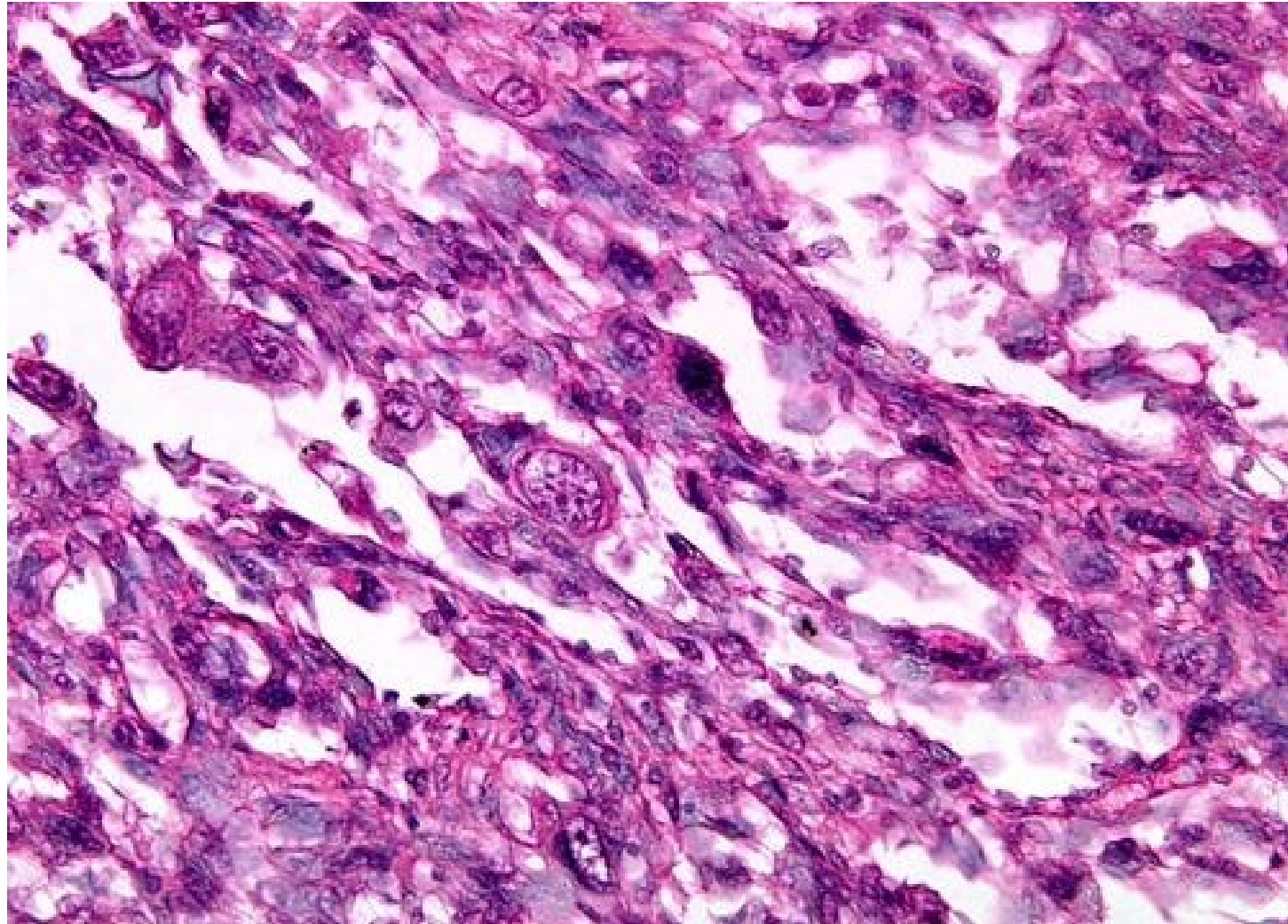
Leiomyosarcoma

- Mesentery, retroperitoneal space, wall of large veins, skin, subcutaneous tissue, deep soft tissues of limbs
- **Signs of malignancy:** large size, high mitotic rate, areas of necrosis, marked cellular pleomorphism

Leiomyoma



Leiomyosarcoma



Cont..

Rhabdomyoma

- Extremely rare lesions

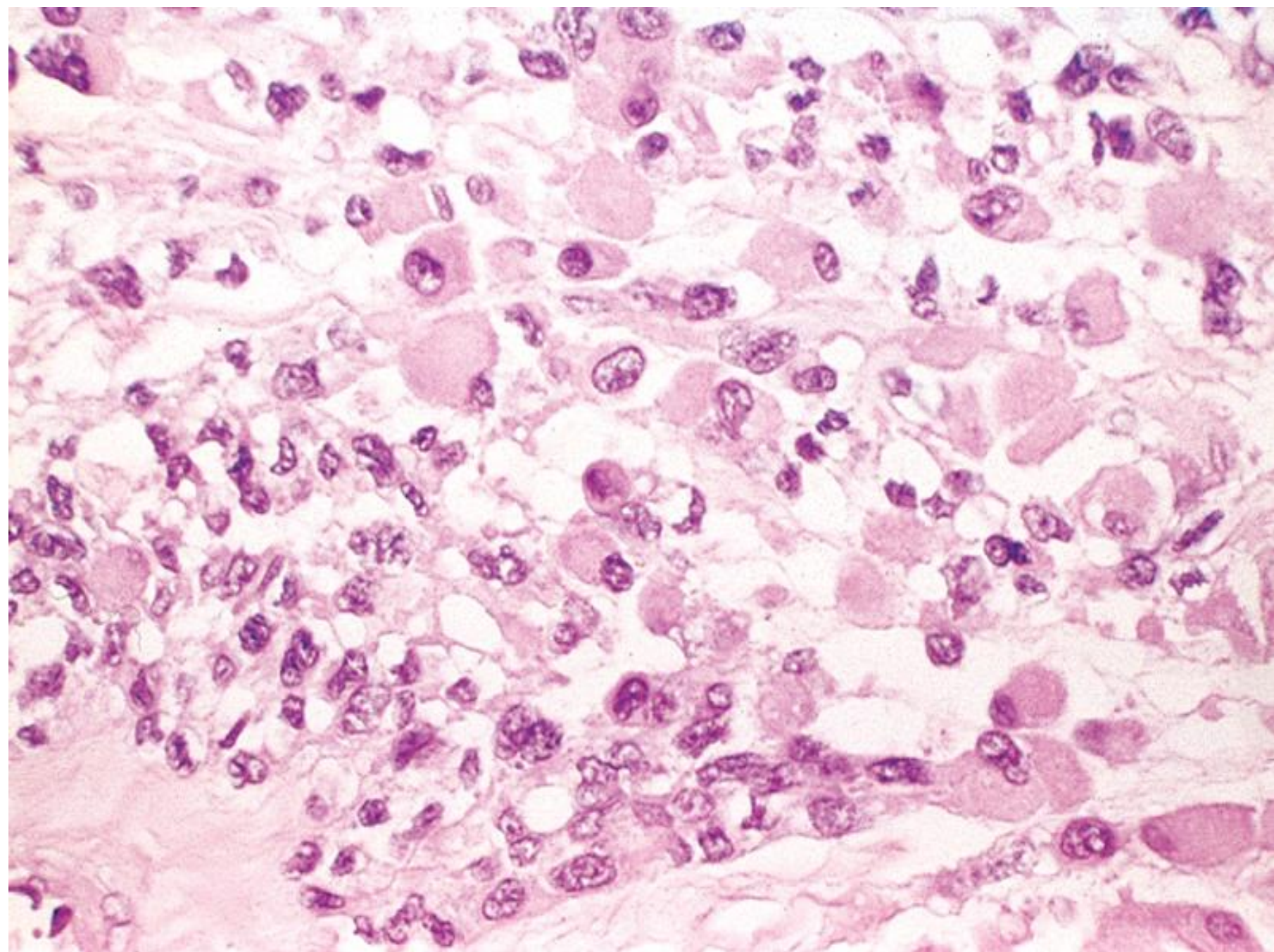
Rhabdomyosarcoma

- The most common malignant soft tissue tumour in infants and young children
- Diagnosis depends on the demonstration of rhabdomyoblasts (round, elongated or oval cells with eccentric eosinophilic cytoplasm, in which fibrillated appearance may be noted – “tadpole cells, strap cells, racket cells“)

Cont..

Subtypes:

- **Embryonal rhabdomyosarcoma**
 - most common, in early childhood, H&N and GU region
 - **small rounded or spindle-shaped** cells in a myxoid matrix
 - **Botryoid rhabdomyosarcoma** (grape-like)
 - Embryonal rhabdomyosarcoma occurring in mucosa-lined organs
- **Alveolar rhabdomyosarcoma**
 - between the ages of 10 to 20 years
 - muscles of limbs and trunk
- **Pleomorphic rhabdomyosarcoma**
 - rare, in limbs of adults,
 - large highly atypical cells seen



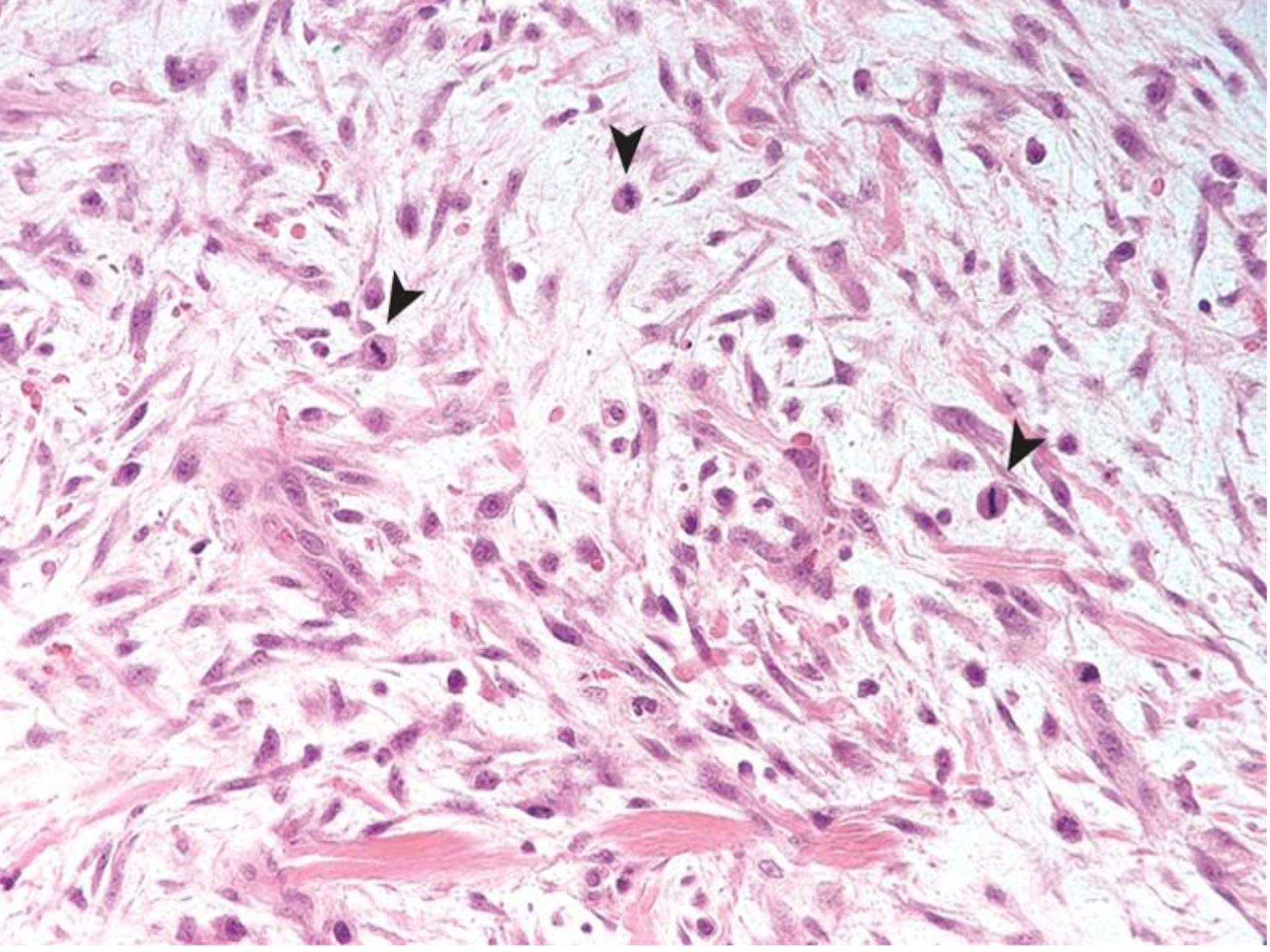
Cont..

Nodular fasciitis ‘ pseudosarcomatous fasciitis’

- Benign reactive fibroblastic proliferation
- Adolescents and young adults
- Rapidly growing nodule within subcutaneous tissue, forearm is the most common site
- Preceding trauma in 10-15% of cases
- Several centimeters , nodular , poorly defined margins

Cont..

- Microscopy-
 - plump immature fibroblasts arranged in randomly or short bundles,
 - numerous mitoses,
 - cellular pleomorphism not present ,
 - lymphocytes and extravasated blood



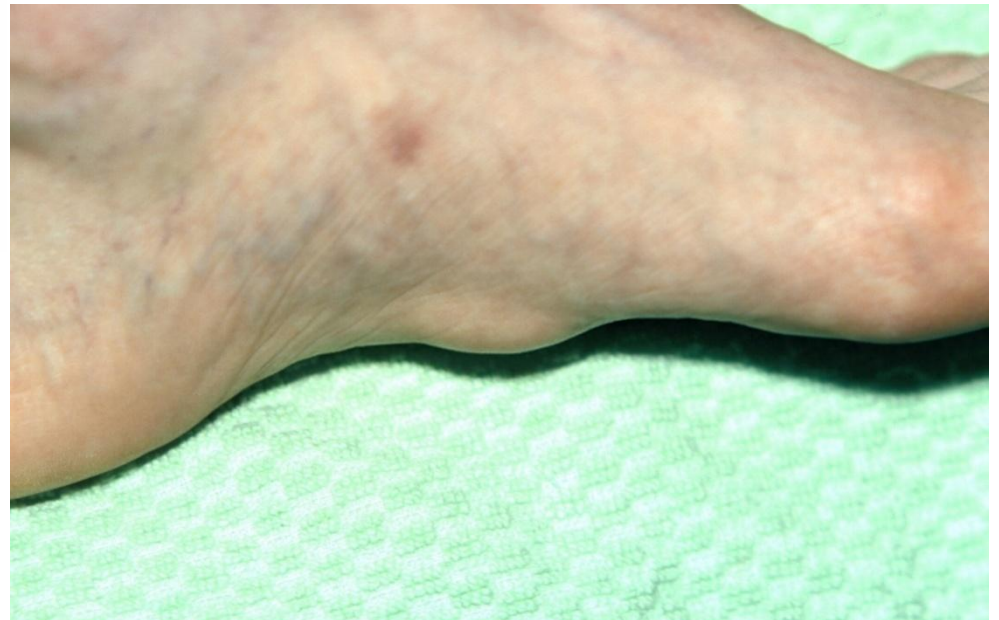
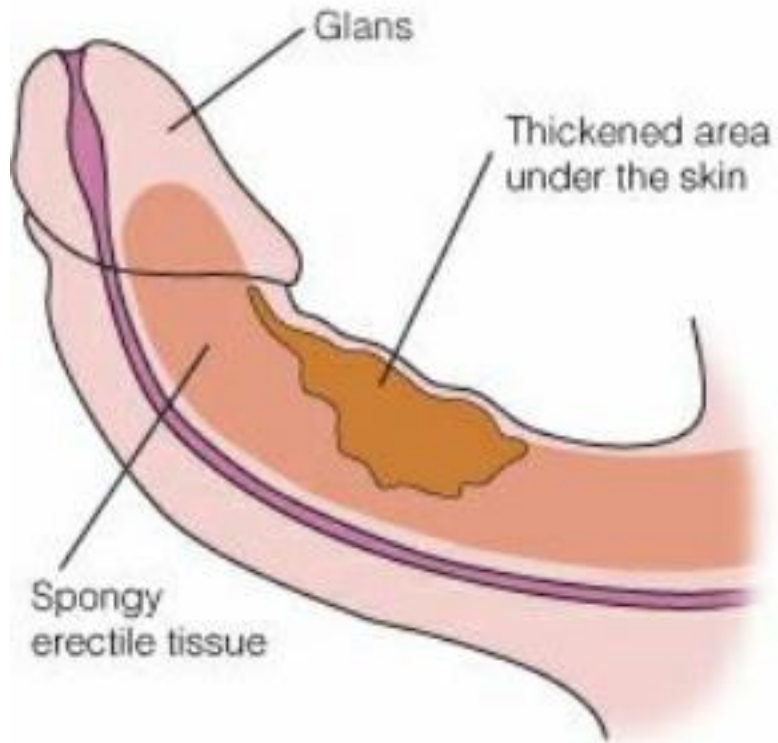
Cont..

Superficial fibromatoses

- **Palmar fibromatosis (Dupuytren's contracture):**
 - nodular thickening of Palmar aponeurosis leading to flexion deformities of fingers
- **Plantar fibromatosis (Ledderhose's disease):**
 - nodular thickening of plantar aponeurosis
- **Penile fibromatosis (Peyronie's disease):**
 - abnormal curvature of penis

Cont..

- All forms are common in males than females
- May have stable course , recur or resolve spontaneously
- Gross and microscopy:
 - irregular margin,
 - nodules of well-differentiated fibroblasts arranged in long sweeping bundles



Cont..

Deep fibromatoses (desmoid tumors)

- **Abdominal:**
 - abdominal wall, young adults, more on women
 - particularly detected in peripartum or postpartum period
 - sometimes in surgical scars
- **Intra-abdominal:**
 - young adults, mesentery,
 - association with Gardner's syndrome (intestinal polyposis)
- **Extra-abdominal:**
 - the most aggressive, adults in the third and fourth decades,
 - pectoral and pelvic girdle

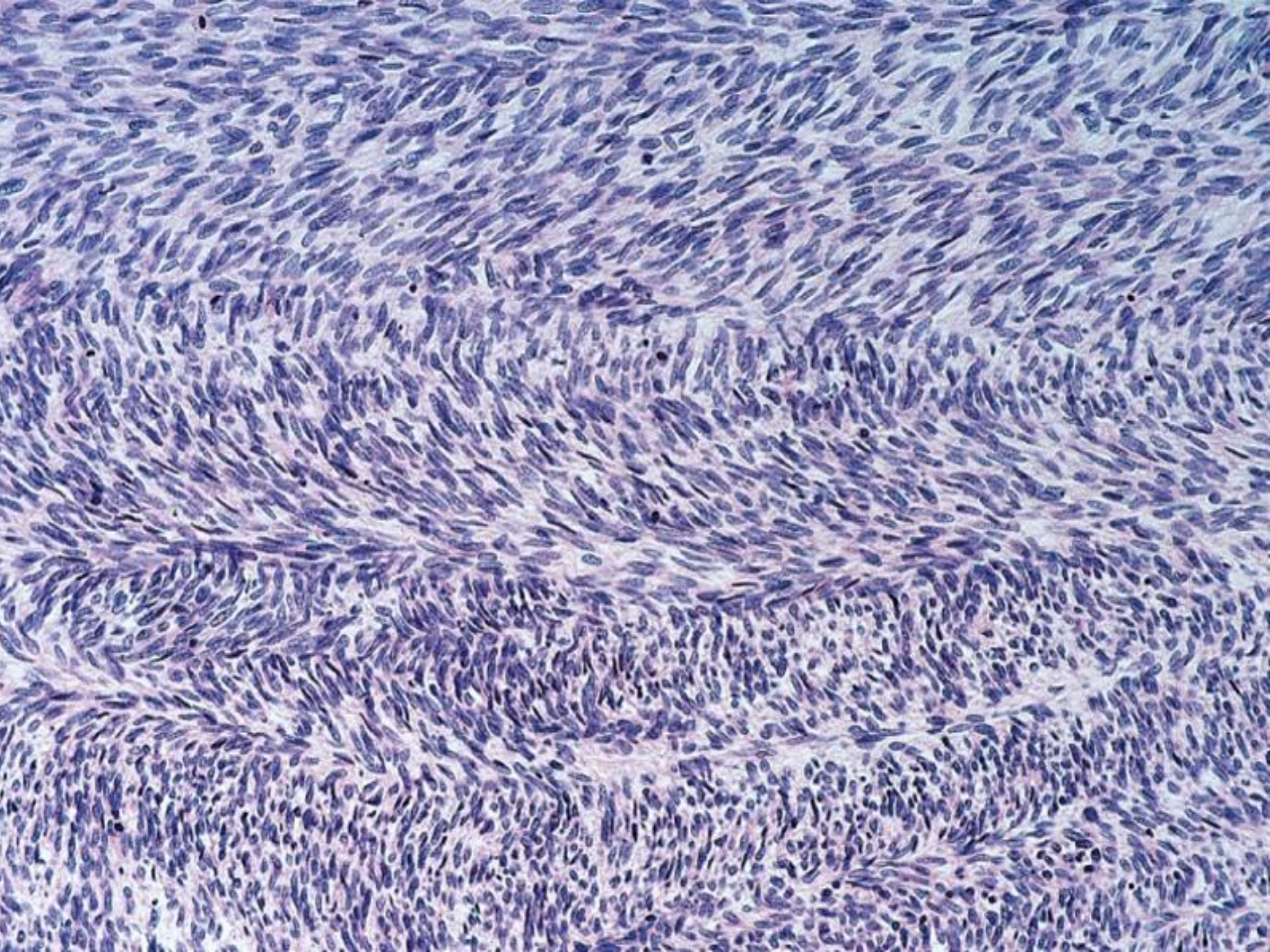
Cont..

- **Features associated with** high risk of recurrence after excision of deep fibromatosis
 - deep intramuscular location,
 - large size (up to 10-15cm),
 - infiltrative growth pattern,

Cont..

Fibrosarcoma

- Relatively uncommon malignant neoplasm
- Middle aged adults
- Deep soft tissues of lower limbs and trunk
- Microscopy: bundles of spindle shaped cells arranged at right angles to one another (“herring-bone pattern”), frequent mitoses
- **Infantile fibrosarcoma:** within the first two years of life, much better prognosis



Cont..

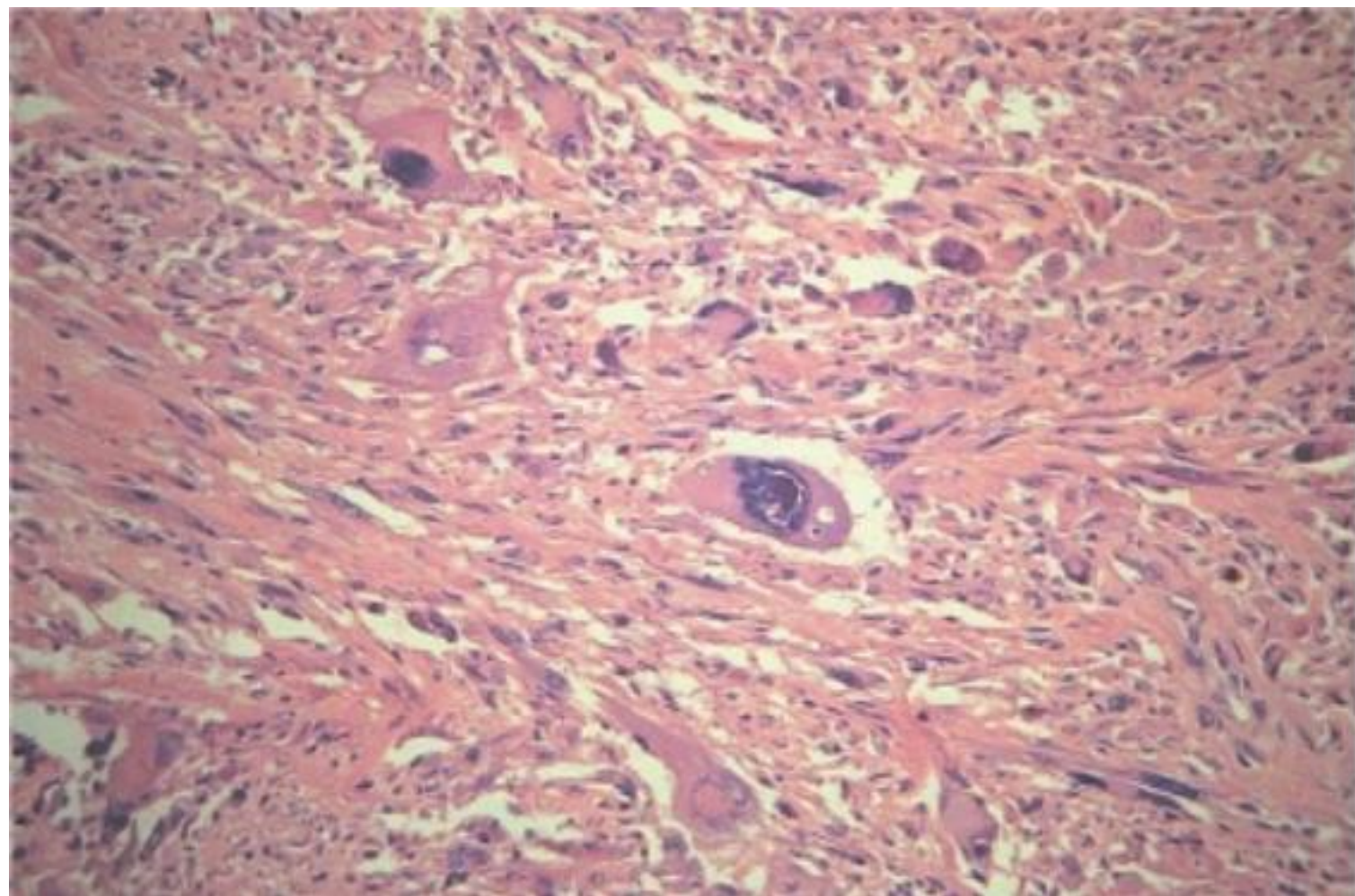
Benign fibrous histiocytoma (dermatofibroma)

- Common lesion, most frequently on the skin of lower leg
- Papule or nodule, often deeply pigmented
- Microscopy: situated within the mid-dermis, spindle cells arranged in curious whorled pattern (storiform pattern)

Cont..

Undifferentiated pleomorphic sarcoma

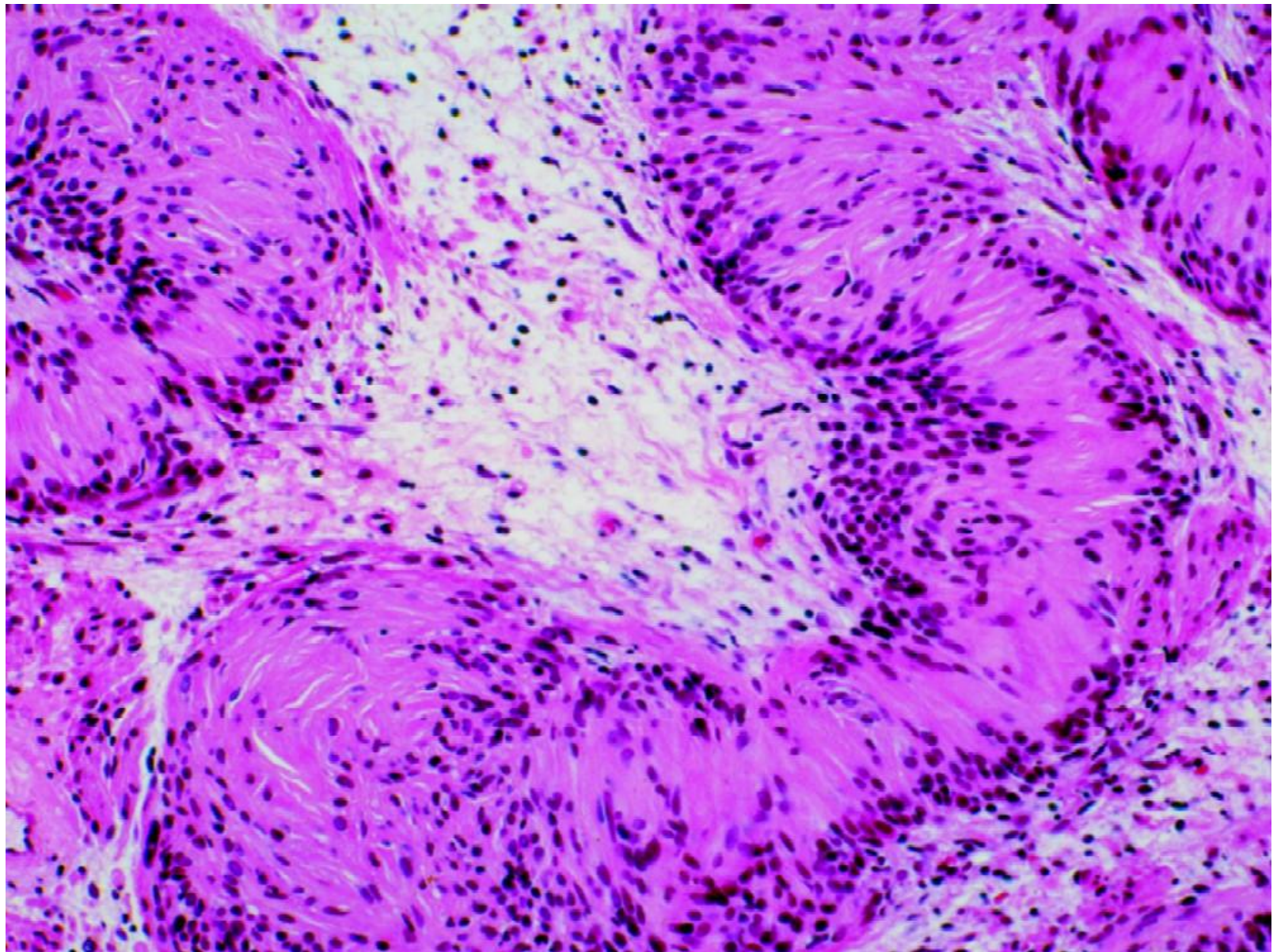
- Deep soft tissues of limbs, retroperitoneum
- Irregularly arranged plump, eosinophilic, spindle-shaped cells ,bizarre nuclei, numerous mitoses, storiform pattern in some areas
- **Ups** represents merely a **morphological pattern** shared by wide variety of **poorly differentiated malignant neoplasms** , it is a heterogeneous group of unrelated lesions
- MFH (synonymous designation: **undifferentiated pleomorphic sarcoma**) - diagnosis of exclusion



Cont..

Schwannoma (neurilemmoma)

- Smooth lobulated lesion usually attached to a nerve
- Gross and microscopy: well circumscribed , two patterns recognized:
- Antoni A – compact areas formed by regular interlacing bundles of uniform spindle-shaped cells, often foci of nuclear palisading
- Antoni B – loose open areas, small cells with rounded nuclei(hypocellular area)



Cont..

Neurofibroma

- Not infrequently multiple, sometimes part of **neurofibromatosis**
- Infiltrates and expands the affected nerve
- Microscopy: spindle-shaped cells with elongated wavy nuclei set in myxoid stroma with scattered mast cells

Cont..

Malignant peripheral nerve sheath tumor (MPNST)

- Adults, most common locations: neck, forearm, lower leg, buttock
- Large mass producing fusiform enlargement of a major nerve
- Microscopy :relatively uniform spindle-shaped cells with hyperchromatic nuclei and high mitotic activity